

Structural Break Estimation and Inference in Temporally Dependent High-Dimensional Dynamic Networks

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Abstract

We study estimation and break-date inference for multiple structural breaks in high-dimensional networks encoded by sparse, regime-specific precision matrices, allowing for temporal dependence in the observations. Global segmentation typically requires repeated high-dimensional network estimation over many candidate intervals, while valid inference demands sharper localization than consistent detection. We propose GRIP-NET, a two-stage procedure that decouples network fitting from optimization over break locations. Starting from a possibly over-specified set of working locations, the procedure estimates the number and preliminary locations of the breaks through a relaxed global optimization based on a limited number of provisional network fits. It then refits the network parameters on the estimated segments and jointly refines all retained break locations through a second global optimization. This “fit few, optimize globally” construction substantially reduces computational cost while retaining global segmentation.

Under a functional-dependence framework for piecewise stationary processes with serial dependence and sub-Weibull tails, we establish consistency for the estimated number of breaks and localization rates for both stages. Global refinement removes the explicit high-dimensional complexity factor and achieves oracle-scale localization. A global-to-local reduction yields limiting distributions for both non-vanishing and vanishing jumps, leading to componentwise and simultaneous confidence intervals. Simulations demonstrate accurate estimation, reliable uncertainty quantification, and substantial computational savings. An application to FRED-MD identifies multiple breaks and substantial reorganization in the contemporaneous dependence structure among U.S. macroeconomic variables.

1 Introduction

Macroeconomic and financial systems are characterized by complex interactions among large collections of variables, making network representations a natural tool for studying their evolving dependence structure. A particularly useful representation is the contemporaneous graphical structure encoded by the precision matrix. Specifically, an edge is present between two variables whenever the corresponding off-diagonal entry of the precision matrix is nonzero. Under Gaussianity, the resulting graph represents conditional independence; More generally, it characterizes contemporaneous linear partial associations among the variables. High-dimensional precision matrices and the associated network structures have therefore been widely used for portfolio allocation, financial risk analysis, and macroeconomic network modeling (e.g., [Chang et al., 2018](#); [Wang et al., 2020](#); [Callot et al., 2021](#); [Chen et al., 2025](#)).

In economic and financial applications, however, such network structures cannot reasonably be expected to remain stable over long observation periods. Changes in monetary and financial conditions, crisis episodes, or broader macroeconomic regimes may alter the contemporaneous relationships among variables. These structural changes may occur repeatedly, making both their number and timing important for understanding how the organization of an economic or financial system evolves across regimes. At the same time, macroeconomic and financial observations typically exhibit serial dependence. Structural-break estimation and break-date inference must therefore accommodate temporal dependence rather than rely on an independence approximation.

To formalize this setting, let $\{\mathbf{X}_i\}_{i=1}^n$ be a sequence of centered p -dimensional observations, allowing temporal dependence within and across segments. Suppose that there exist an unknown number of change points $0 = \tau_0^* < \tau_1^* < \dots < \tau_{k^*}^* < \tau_{k^*+1}^* = n$ such that

$$\text{Cov}(\mathbf{X}_i) = \boldsymbol{\Sigma}_j^*, \quad \boldsymbol{\Omega}_j^* = (\boldsymbol{\Sigma}_j^*)^{-1}, \quad \tau_{j-1}^* < i \leq \tau_j^*, \quad j \in [k^* + 1]. \quad (1)$$

For each segment, the sparse positive-definite precision matrix $\boldsymbol{\Omega}_j^*$ determines a graph $G_j^* = (V, E_j^*)$, where $V = [p]$ and $E_j^* = \left\{ (a, b) : 1 \leq a < b \leq p, \omega_{j,ab}^* \neq 0 \right\}$. A change point therefore represents a structural change in the contemporaneous precision matrix-based network, with $\boldsymbol{\Omega}_j^* \neq \boldsymbol{\Omega}_{j+1}^*$. Our objective is to estimate the unknown number and locations of these structural breaks, recover the corresponding regime-specific network structures, and quantify uncertainty about the estimated break dates while allowing for temporal dependence.

The structural object considered here is distinct from several other forms of time-varying dependence studied in the econometric literature. [Barigozzi et al. \(2018\)](#), for example, consider multiple changes in the general second-order structure of high-dimensional factor time series and distinguish changes in the common and idiosyncratic components. [Bai et al. \(2020\)](#) study

estimation and inference for a break in high-dimensional factor loadings. In high-dimensional vector autoregressions, [Safikhani and Shojaie \(2022\)](#) focus on changes in lagged predictive relationships represented by transition matrices, whereas [Casini and Perron \(2024\)](#) investigate structural changes in frequency-domain dependence through evolutionary spectra. These studies concern changes in factor structures, loadings, lagged transition coefficients, or frequency-domain dependence. In contrast, we study abrupt structural changes in the contemporaneous precision matrix itself. The resulting breaks capture changes in direct linear relationships after controlling for the remaining variables and, under Gaussianity, correspond to changes in the conditional-independence network. They therefore represent a different form of structural instability from changes in latent factor structure, predictive dynamics, or frequency-domain dependence.

The main difficulty is that multiple-break estimation, high-dimensional network estimation, and break-date inference are tightly coupled. Estimating a high-dimensional precision-matrix-based network, for example through sparse nodewise regressions, is already computationally demanding. Global estimation of multiple break locations further requires comparisons over a large collection of candidate segmentations. Because the network parameters are unknown and segment-specific, evaluating these candidate segmentations ordinarily requires repeated high-dimensional network estimation, creating a severe computational bottleneck. Moreover, valid break-date inference requires substantially sharper localization than consistent detection alone. These considerations give rise to two related challenges: reducing the computational burden of global segmentation and attaining the sharper localization required for break-date inference.

Existing change-point procedures seek to mitigate the first challenge through fused regularization, binary segmentation, divide-and-conquer strategies, and other approximate or restricted search schemes (e.g., [Kolar and Xing, 2012](#); [Gibberd and Nelson, 2017](#); [Bybee and Atchadé, 2018](#); [Londschien et al., 2021](#); [Liu et al., 2021](#); [Li et al., 2023](#)). Related scalable procedures have also been developed for more general high-dimensional change-point problems ([Bai and Safikhani, 2023a](#); [Kovács et al., 2024a](#)). These approaches can substantially reduce the computational burden of high-dimensional segmentation, but they do not resolve the basic computational tension arising in models with expensive nuisance estimation. In such settings, the dominant cost is often not the optimization over candidate break locations itself, but the repeated high-dimensional model fitting required to evaluate those candidates. This distinction suggests a different computational principle: estimate the expensive network parameters only a limited number of times while retaining global optimization over the break locations. Such a separation provides a route to computationally feasible global segmentation, which can then serve as the basis for the sharper refinement required for subsequent inference.

Beyond accurately estimating the structural breaks, empirical interpretation requires quantifying uncertainty about their locations. A point estimate alone does not reveal how precisely

a break has been localized or how confidently its timing can be associated with an external economic event. Break-date inference has a long history in econometric structural-change models, including regression and high-dimensional factor models (e.g., [Bai, 1997](#); [Bai and Perron, 1998, 2003](#); [Bai et al., 2020](#)). More recently, confidence intervals for change-point locations have been developed for several high-dimensional mean, regression, and covariance change models (e.g., [Chen et al., 2022](#); [Kaul et al., 2021](#); [Zhang and Kaul, 2024](#); [Xu et al., 2024](#)). For high-dimensional precision matrix based-networks, however, the structures on both sides of each break are unknown and must themselves be estimated. Existing work has therefore focused primarily on detection consistency, localization rates, and network estimation, while inferential results remain limited and mainly concern a single graphical change point ([Kaul et al., 2023](#)).

With multiple structural breaks, break-date inference introduces an additional layer of difficulty. A preliminary procedure must first consistently recover the number of breaks and localize each break within a vanishing fraction of the adjacent segment lengths. This level of accuracy may be sufficient for consistent detection but remains too coarse for break-date inference. A refinement mechanism is therefore needed to move beyond the initial high-dimensional localization rate and deliver break estimators at the sharper scale required for limiting distribution theory and confidence-interval construction.

Temporal dependence further affects every stage of this inferential problem. Most existing graphical change-point theory assumes independent observations, an approximation that is restrictive for macroeconomic and financial time series. Serial dependence influences the estimation of regime-specific precision matrix based-networks, the stochastic fluctuations governing break-location errors, and hence the limiting distributions and calibration of confidence intervals. Precision-matrix estimation through nodewise methods has been studied for stationary dependent sequences without structural breaks ([Chang et al., 2018](#); [Callot et al., 2021](#)), while graphical representations of stationary time series have also been investigated in the frequency domain ([Dahlhaus, 2000](#); [Tugnait, 2022](#); [Krampe and Paparoditis, 2025](#)). These frameworks, however, do not provide joint estimation and break-date inference for multiple abrupt changes in a high-dimensional contemporaneous precision matrix based-network.

The problem therefore calls for an integrated framework that combines computationally feasible global segmentation, statistically sharp localization through refinement, and multiple-break inference under temporal dependence. The methodology developed below is designed precisely around this estimation–refinement–inference sequence. Our main contributions are summarized as follows.

First, we propose GRIP-NET, short for *Global Refinement and Inference for Structural Breaks in High-Dimensional precision matrix based-networks*, an estimation–refinement framework that decouples expensive high-dimensional network estimation from global optimization

over break locations. Starting from a possibly over-specified collection of working locations, the method fits provisional network parameters only on the induced intervals and, with these parameters held fixed, performs a relaxed global optimization that removes redundant candidates and estimates the preliminary segmentation. The network parameters are then refitted on the estimated segments, after which all retained break locations are jointly refined through a second global optimization. This “fit few, optimize globally” architecture retains global segmentation criteria while reducing the number of expensive precision-network fits from $O(n^2)$ candidate-interval fits in conventional global segmentation to $O(\tilde{k} + \hat{k})$.

Second, we show that this computational decoupling also produces a substantial statistical sharpening. The preliminary global reassignment consistently estimates the number of structural breaks and satisfies

$$|\hat{\tau}_j - \tau_j^*| = O_{\mathbb{P}} \left\{ \frac{r_{\zeta}(n, p)}{\kappa_j^2} \right\}, \quad r_{\zeta}(n, p) = \{s \log(np)\}^{2/\zeta-1},$$

where the complexity factor reflects network sparsity, dimensionality, temporal dependence, and tail behavior. After segment-wise refitting, global refinement removes this explicit high-dimensional factor: the refined estimator attains $O_{\mathbb{P}}(\kappa_j^{-2})$ localization for vanishing changes and $O_{\mathbb{P}}(1)$ localization for non-vanishing changes. Thus, refinement is not merely a numerical adjustment to the preliminary segmentation and it moves the estimator from a high-dimensional localization scale to the oracle scale required for break-date inference.

Third, we develop break-date inference for multiple structural changes in high-dimensional precision matrix based-networks. Although the refined estimator is defined through a joint global optimization over all break locations, its asymptotic behavior admits a global-to-local reduction: around each true break, the relevant part of the global criterion is asymptotically equivalent to a local single-break contrast problem. This reduction provides the key bridge from globally defined multiple-break estimation to tractable inference. For non-vanishing changes, the unscaled localization error converges to the maximizer of a two-sided random walk, whereas for vanishing changes the properly normalized error converges to the maximizer of a two-sided Brownian motion with linear drift. These limits yield asymptotically valid componentwise confidence intervals. We further establish joint convergence and asymptotic independence of the local limiting processes associated with distinct break points, enabling simultaneous confidence intervals for a fixed number of structural breaks.

Fourth, the entire estimation–refinement–inference theory is developed under a unified functional-dependence framework (Wu, 2011). The observations may form piecewise stationary processes with serial dependence and sub-Weibull tail behavior, with independent observations included as a special case. Temporal dependence enters both the concentration analysis underlying high-

dimensional precision matrix estimation and preliminary localization and the stochastic fluctuations governing the refined break estimators and their limiting distributions. The dependent framework is therefore integrated throughout the methodology rather than appended as a robustness extension, making the results directly relevant to macroeconomic and financial time series.

The numerical and empirical analyses provide complementary evidence for these methodological and theoretical results. Simulation experiments examine estimation of the number and locations of structural breaks, the localization improvement produced by global refinement, componentwise and simultaneous confidence-interval coverage, and computational performance under both independent and serially dependent observations. The results show that GRIP-NET combines accurate segmentation and reliable uncertainty quantification with substantial computational savings relative to competing procedures. In an application to the FRED-MD macroeconomic database, the method identifies multiple structural changes in the contemporaneous precision matrix based-network, including breaks temporally associated with the 1973–1975 recession and recovery, the global financial crisis, and the COVID-19 recession. Regime-specific network estimates further reveal substantial reorganization in both within-category and cross-category macroeconomic linkages across the estimated regimes.

The remainder of the paper is organized as follows. Section 2 introduces the multiple-change-point graphical model and the proposed estimation and global-refinement procedure. Section 3 establishes the theoretical properties of the preliminary and refined estimators and develops the limiting distributions and confidence intervals. Section 4 reports the simulation results, and Section 5 presents the FRED-MD application. Additional technical results, proofs, numerical results, and implementation details are provided in the Supplementary Material.

2 Methodology

This section introduces the nodewise representation of high-dimensional precision matrix based-networks, the segment loss used for structural-break localization, and the proposed GRIP-NET procedure. We then develop the computational motivation, global refinement, and inference construction.

2.1 Nodewise Representation and Segment Loss

Let $\{\mathbf{X}_i\}_{i=1}^n$ be a sequence of centered p -dimensional random vectors, where $\mathbf{X}_i = (X_{i,1}, \dots, X_{i,p})^\top \in \mathbb{R}^p$.

The observations are allowed to be temporally dependent and are not required to be Gaussian. Suppose that there exist an unknown number of change points $\mathcal{T}_{k*}^* = \{\tau_1^*, \dots, \tau_{k*}^*\}$, with $0 =$

$\tau_0^* < \tau_1^* < \dots < \tau_{k^*}^* < \tau_{k^*+1}^* = n$, such that the contemporaneous covariance matrix is constant within each segment:

$$\mathbb{E}(\mathbf{X}_i \mathbf{X}_i^\top) = \mathbf{\Sigma}_j^*, \quad \tau_{j-1}^* < i \leq \tau_j^*, \quad j \in [k^* + 1],$$

where each $\mathbf{\Sigma}_j^* \in \mathbb{R}^{p \times p}$ is positive definite. Conditions governing the temporal dependence and tail behavior of $\{\mathbf{X}_i\}_{i=1}^n$ are introduced in Section 3. Denote the segment-specific precision matrix by

$$\mathbf{\Omega}_j^* = (\mathbf{\Sigma}_j^*)^{-1} = (\omega_{j,ab}^*)_{1 \leq a, b \leq p}.$$

For each segment $j \in [k^* + 1]$, the precision matrix $\mathbf{\Omega}_j^*$ defines a contemporaneous precision matrix based-network $G_j^* = (V, E_j^*)$, with $V = [p]$ and edge set

$$E_j^* = \{(a, b) : 1 \leq a < b \leq p, \omega_{j,ab}^* \neq 0\}.$$

In general, this network represents contemporaneous linear partial dependence among the components of $\mathbf{X}_i$. Under Gaussianity, a zero off-diagonal entry of $\mathbf{\Omega}_j^*$ additionally corresponds to conditional independence, so that the precision matrix based-network coincides with the usual Gaussian graphical model. A change point therefore represents a structural change in the contemporaneous precision matrix based-network, with $\mathbf{\Omega}_j^* \neq \mathbf{\Omega}_{j+1}^*$.

For each segment $j \in [k^* + 1]$ and node $a \in [p]$, define the segment-specific population nodewise regression coefficient as $\boldsymbol{\theta}_j^{a*} = (\theta_{j,1}^{a*}, \dots, \theta_{j,p}^{a*})^\top$ with

$$\boldsymbol{\theta}_j^{a*} = \arg \min_{\substack{\boldsymbol{\theta} \in \mathbb{R}^p \\ \theta_a = 0}} \frac{1}{\tau_j^* - \tau_{j-1}^*} \mathbb{E} \left[\sum_{i=\tau_{j-1}^*+1}^{\tau_j^*} \left(X_{i,a} - \mathbf{X}_i^\top \boldsymbol{\theta} \right)^2 \right]. \quad (2)$$

Collecting the p nodewise regression coefficient vectors gives the segment-specific coefficient matrix in the j th segment,

$$\boldsymbol{\Theta}_j^* = \begin{pmatrix} (\boldsymbol{\theta}_j^{1*})^\top \\ (\boldsymbol{\theta}_j^{2*})^\top \\ \vdots \\ (\boldsymbol{\theta}_j^{p*})^\top \end{pmatrix} = \begin{pmatrix} 0 & \theta_{j,2}^{1*} & \dots & \theta_{j,p}^{1*} \\ \theta_{j,1}^{2*} & 0 & \dots & \theta_{j,p}^{2*} \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{j,1}^{p*} & \theta_{j,2}^{p*} & \dots & 0 \end{pmatrix}, \quad j \in [k^* + 1]. \quad (2)$$

Note that by construction, the a th entry of $\boldsymbol{\theta}_j^{a*}$ is zero.

The coefficient vector $\boldsymbol{\theta}_j^{a*}$ characterizes the contemporaneous linear projection of node a on the remaining components of $\mathbf{X}_i$. By the normal equations, for every $a \neq b$, (see, e.g., [Lauritzen](#),

1996),

$$\theta_{j,b}^{a*} = -\frac{\omega_{j,ab}^*}{\omega_{j,aa}^*}.$$

Consequently, the zero pattern of the nodewise coefficient matrix Θ_j^* coincides with the zero pattern of the off-diagonal entries of the precision matrix Ω_j^* . Thus, Θ_j^* provides an equivalent representation of the contemporaneous precision network used in our estimation procedure. Under Gaussianity, this zero pattern also corresponds to the conditional-independence graph.

The population quantities above characterize the network structure within each true segment. In practice, however, neither the true break locations nor the corresponding nodewise coefficient matrices are known. A structural-break procedure must therefore estimate a coefficient matrix on arbitrary candidate intervals and evaluate how well collections of such interval-specific parameters fit the full sequence. We next introduce the empirical segment loss and the associated nodewise estimators used to construct the global segmentation criteria.

For any node $a \in [p]$, integer interval $(u, v]$ with $0 \leq u < v \leq n$, and coefficient vector $\theta = (\theta_1, \dots, \theta_p)^\top \in \mathbb{R}^p$, define the nodewise segment loss by

$$L^a(u, v; \theta) = \sum_{i=u+1}^v \left(X_{i,a} - \mathbf{X}_i^\top \theta \right)^2.$$

For any matrix $\Theta = (\theta^1, \dots, \theta^p)^\top \in \mathbb{R}^{p \times p}$ with zero diagonal, define the aggregated segment loss

$$\mathcal{L}(u, v; \Theta) = \begin{cases} \frac{1}{p} \sum_{a=1}^p L^a(u, v; \theta^a), & u < v, \\ 0, & u = v. \end{cases} \quad (3)$$

For a tuning parameter $\lambda > 0$, the corresponding nodewise Lasso estimator for θ_j^{a*} is

$$\hat{\theta}_{(u,v]}^{a,\lambda} = \arg \min_{\substack{\theta \in \mathbb{R}^p \\ \theta_a = 0}} \left\{ L^a(u, v; \theta) + \lambda \sqrt{v-u} \|\theta\|_1 \right\}. \quad (4)$$

Collecting the p nodewise estimators gives the matrix estimator corresponding to Θ_j^* :

$$\hat{\Theta}_{(u,v]}^\lambda = \left((\hat{\theta}_{(u,v]}^{1,\lambda})^\top, \dots, (\hat{\theta}_{(u,v]}^{p,\lambda})^\top \right)^\top \in \mathbb{R}^{p \times p}. \quad (5)$$

To combine the aggregated segment losses defined above into a global segmentation criterion, we next specify the admissible change-point configurations. For $v \in [n]$ and $k \geq 0$, define the strict change-point space by

$$\mathbb{T}_k(v) = \left\{ \tau = (\tau_1, \dots, \tau_k)^\top : 0 = \tau_0 < \tau_1 < \dots < \tau_k < \tau_{k+1} = v \right\}.$$

When $v = n$, we write $\mathbb{T}_k = \mathbb{T}_k(n)$. We further define the relaxed change-point space

$$\bar{\mathbb{T}}_k(v) = \left\{ \mathbf{u} = (u_1, \dots, u_k)^\top : 0 = u_0 < u_1 \leq \dots \leq u_k \leq u_{k+1} = v \right\}.$$

Unlike $\mathbb{T}_k(v)$, the relaxed space $\bar{\mathbb{T}}_k(v)$ permits repeated adjacent locations. If $u_j = u_{j+1}$, the corresponding interval $(u_j, u_{j+1}]$ has zero length and contributes zero to the aggregate loss in (3). This allows redundant guessed segments to be collapsed in the preliminary optimization.

2.2 Motivation and Overview of GRIP-Net

The estimation problem is challenging because the number of structural breaks k^* , their locations $\mathcal{T}_{k^*}$, and the regime-specific nodewise coefficient matrices $\{\Theta_j^*\}_{j=1}^{k^*+1}$ are all unknown. A natural benchmark is exact global segmentation by dynamic programming. Specifically, for a candidate segmentation $\boldsymbol{\tau} = (\tau_1, \dots, \tau_k)^\top \in \mathbb{T}_k(n)$, one may estimate a separate coefficient matrix on each segment $(\tau_{j-1}, \tau_j]$ and solve

$$\min_{k \geq 0} \min_{\boldsymbol{\tau} \in \mathbb{T}_k(n)} \left\{ \sum_{j=1}^{k+1} \mathcal{L}(\tau_{j-1}, \tau_j; \hat{\Theta}_{(\tau_{j-1}, \tau_j]}^\lambda) + \gamma(k+1) \right\}, \quad (6)$$

where $\gamma > 0$ penalizes the number of fitted segments and $\hat{\Theta}_{(\tau_{j-1}, \tau_j]}^\lambda$ is estimated separately on each candidate segment. Owing to the additive structure of the criterion, (6) can be solved exactly by dynamic programming (Jackson et al., 2005).

The main computational bottleneck, however, is not the dynamic programming recursion itself but the construction of the segment costs in (6). There are $O(n^2)$ candidate intervals, and evaluating the criterion on each interval requires a separate high-dimensional nodewise fit. If $\mathbf{Graph}(n, p)$ denotes an upper bound on the cost of estimating one precision-network coefficient matrix, direct global segmentation therefore has computational cost $O\{n^2 \mathbf{Graph}(n, p)\}$. Under the nodewise regression formulation, $\mathbf{Graph}(n, p) = O\{p \cdot \mathbf{Lasso}(n, p)\}$, so repeated high-dimensional fitting over all candidate intervals can quickly dominate the cost of the segmentation problem.

Binary segmentation provides a computationally cheaper alternative (Fryzlewicz, 2014). Rather than optimizing jointly over all break locations, it recursively searches for one break within each current segment. Under the usual balanced-recursion accounting, this requires $O\{n \log(n)\}$ candidate evaluations and hence has cost $O\{n \log(n) \mathbf{Graph}(n, p)\}$. Although substantially cheaper than exact dynamic programming, it can still require many expensive high-dimensional fits and, unlike a global segmentation criterion, optimizes the break locations recursively rather than jointly.

These observations suggest separating the expensive estimation of the high-dimensional nui-

sance parameters from the optimization over break locations. We therefore propose GRIP-NET, short for *Global Refinement and Inference for Structural Breaks in High-Dimensional precision matrix based-networks*. The method follows a “fit few, optimize globally” principle: the network parameters are estimated only on a limited collection of intervals, while the break locations are optimized globally conditional on these fitted parameters. This decoupling reduces the number of expensive high-dimensional fits while retaining global segmentation criteria. It also provides the refinement needed to move beyond the preliminary high-dimensional localization scale to the sharper scale required for break-date inference.

In Step 1, an over-specified collection of working break locations is used to construct provisional nodewise coefficient matrices. Conditional on these matrices, the locations are globally reassigned over a relaxed change-point space, simultaneously eliminating redundant candidates and producing preliminary estimates of both the number and locations of the structural breaks. The provisional matrices are required only to represent the underlying regimes sufficiently well, so the resulting preliminary locations need not yet attain the localization scale required for inference. In Step 2, the coefficient matrices are refitted on the preliminary segmentation. Holding these refitted matrices fixed, all retained break locations are then jointly refined through a second global optimization. The resulting estimators attain the sharper localization rates underlying the limiting distribution theory and confidence-interval construction developed below; see Figure 1.

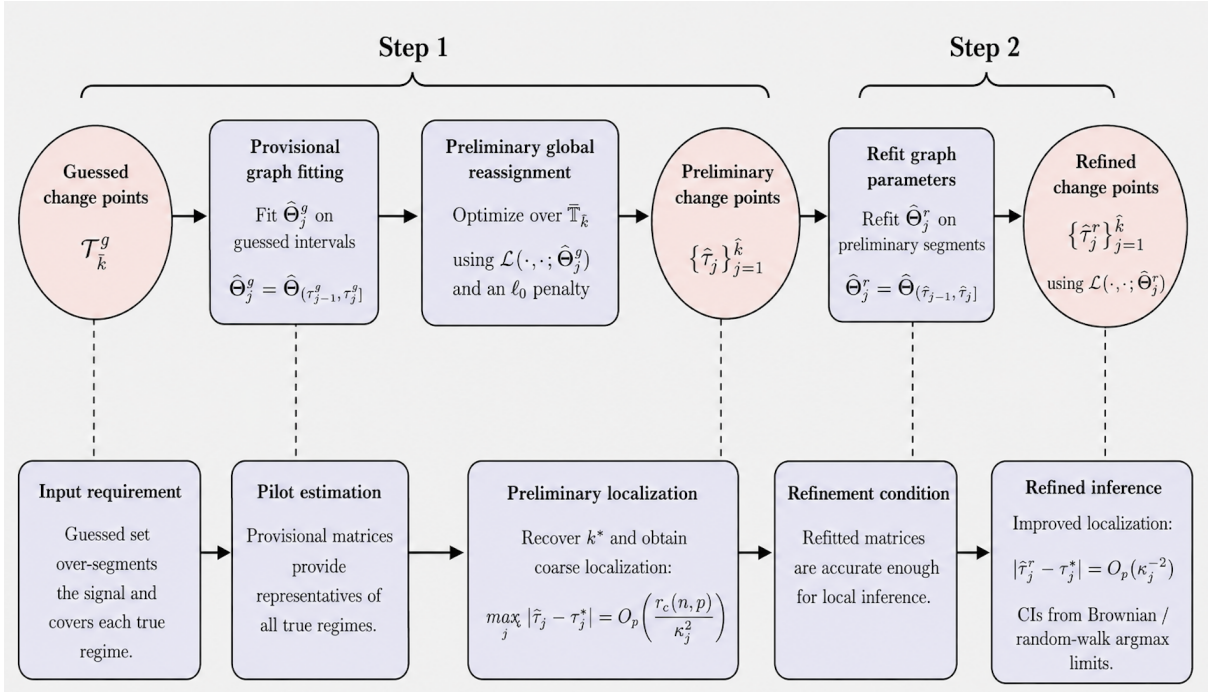


Figure 1: A schematic of the two-stage GRIP-NET procedure and its estimation–refinement–inference mechanism.

2.3 GRIP-NET

We now present the two stages of GRIP-NET in detail.

Step 1: preliminary global reassignment. Here and throughout, the superscript g indicates quantities constructed from the guessed change-point locations. Let

$$\mathcal{T}_{\tilde{k}}^g = \{\tau_1^g, \dots, \tau_{\tilde{k}}^g\} \in \mathbb{T}_{\tilde{k}}, \quad \tilde{k} \geq k^*,$$

be an over-segmenting collection of working change-point locations, where $\tau_0^g = 0$ and $\tau_{\tilde{k}+1}^g = n$. These locations may be obtained from an initial screening procedure or supplied through a sufficiently dense deterministic grid. Their role is only to provide a coarse partition from which provisional regime-specific parameters can be estimated; they are not treated as final break estimators.

Applying (4) and (5) to the resulting $\tilde{k} + 1$ intervals with tuning parameter λ^g yields the ordered collection of provisional coefficient matrices

$$\hat{\Theta}_{[\tilde{k}+1]}^g = \{\hat{\Theta}_j^g\}_{j=1}^{\tilde{k}+1} = \left\{ \hat{\Theta}_{(\tau_0^g, \tau_1^g]}^{\lambda^g}, \hat{\Theta}_{(\tau_1^g, \tau_2^g]}^{\lambda^g}, \dots, \hat{\Theta}_{(\tau_{\tilde{k}}^g, \tau_{\tilde{k}+1}^g]}^{\lambda^g} \right\}. \quad (7)$$

Here $\lambda^g > 0$ is the nodewise Lasso tuning parameter used in the provisional stage. The working locations serve only to generate a collection of provisional parameter estimates sufficiently rich to represent all underlying regimes. Informally, for each true segment $j \in [k^* + 1]$, there should exist at least one $j \in [\tilde{k} + 1]$ such that $\|\hat{\Theta}_j^g - \Theta_j^*\|_F$ is sufficiently small. The precise approximation requirement is formalized in Assumption 4. Importantly, this condition does not require the working locations themselves to be accurately localized: $\mathcal{T}_{\tilde{k}}^g$ may contain redundant locations, and several provisional matrices may represent the same underlying regime.

Conditional on $\{\hat{\Theta}_j^g\}_{j=1}^{\tilde{k}+1}$, the break locations are reassigned by solving the following ℓ_0 -penalized global problem:

$$\hat{\mathcal{U}}_{\tilde{k}} := (\hat{u}_1, \dots, \hat{u}_{\tilde{k}}) = \arg \min_{\mathcal{U} \in \bar{\mathbb{T}}_{\tilde{k}}(n)} \left\{ \sum_{j=1}^{\tilde{k}+1} \mathcal{L}(u_{j-1}, u_j; \hat{\Theta}_j^g) + \gamma \sum_{j=1}^{\tilde{k}+1} \mathbf{1}_{\{u_{j-1} < u_j\}} \right\}, \quad (8)$$

where $\mathcal{U} = (u_1, \dots, u_{\tilde{k}})^\top$, $u_0 = 0$, $u_{\tilde{k}+1} = n$, and $\gamma > 0$ controls the number of nonempty fitted intervals. Because $\bar{\mathbb{T}}_{\tilde{k}}(n)$ permits repeated adjacent locations, the penalty counts the effective number of fitted segments rather than the nominal number $\tilde{k} + 1$. Redundant provisional regimes can therefore be assigned zero-length intervals.

Let $\hat{\mathcal{U}}_{\tilde{k}} = (\hat{u}_1, \dots, \hat{u}_{\tilde{k}})^\top$ solve (8). The preliminary change-point set is obtained by removing repeated locations:

$$\widehat{\mathcal{T}}_{\hat{k}} = \text{Uni}(\widehat{\mathcal{U}}_{\hat{k}}) := \{\hat{u}_j : \hat{u}_j \neq \hat{u}_{j+1}, j \in [\hat{k}]\}. \quad (9)$$

The estimated number of structural breaks is then $\hat{k} = |\widehat{\mathcal{T}}_{\hat{k}}|$.

The optimization in (8) thus retains provisional coefficient matrices that improve the global fit while collapsing redundant ones through zero-length intervals. Under the conditions introduced below, the resulting preliminary estimator consistently recovers the number of structural breaks and satisfies $\max_{j \in [k^*]} \kappa_j^2 \left| \hat{\tau}_j - \tau_j^* \right| = O_p\{r_\zeta(n, p)\}$, where κ_j denotes the signal magnitude at the j th structural break and $r_\zeta(n, p)$ captures the high-dimensional complexity arising from sparsity, dimensionality, temporal dependence, and tail behavior. The refinement step below removes this explicit complexity factor, yielding $O_p(\kappa_j^{-2})$ localization for vanishing jumps and $O_p(1)$ localization for non-vanishing jumps.

Step 2-1: global refinement. The preliminary estimators $\widehat{\mathcal{T}}_{\hat{k}} = (\hat{\tau}_1, \dots, \hat{\tau}_{\hat{k}})^\top$, with $\hat{\tau}_0 = 0$ and $\hat{\tau}_{\hat{k}+1} = n$, partition the sample into $\hat{k} + 1$ intervals.

We refit the nodewise coefficient matrices on these intervals and obtain

$$\widehat{\Theta}_{[\hat{k}+1]}^r = \left(\widehat{\Theta}_1^r, \dots, \widehat{\Theta}_{\hat{k}+1}^r \right), \quad (10)$$

where $\widehat{\Theta}_j^r = \widehat{\Theta}_{(\hat{\tau}_{j-1}, \hat{\tau}_j]}^{\lambda^r}, j \in [\hat{k} + 1]$. The superscript r denotes quantities obtained after refitting, and $\lambda^r > 0$ is the corresponding nodewise Lasso tuning parameter, which may differ from the provisional-stage parameter λ^g .

Holding the refitted coefficient matrices fixed, we define the refined break estimators through the joint global optimization

$$\widehat{\mathcal{T}}_{\hat{k}}^r := (\hat{\tau}_1^r, \dots, \hat{\tau}_{\hat{k}}^r)^\top \in \arg \min_{\mathcal{T} \in \mathbb{T}_{\hat{k}}(n)} \sum_{j=1}^{\hat{k}+1} \mathcal{L}(\tau_{j-1}, \tau_j; \widehat{\Theta}_j^r). \quad (11)$$

The optimization is now performed over the strict space $\mathbb{T}_{\hat{k}}(n)$ because the number of breaks has already been selected in Step 1 and every retained regime is required to be nonempty. Hence, no additional model-selection penalty is needed in (11).

The refinement criterion is motivated by the oracle problem

$$\arg \min_{\mathcal{T} \in \mathbb{T}_{k^*}(n)} \sum_{j=1}^{k^*+1} \mathcal{L}(\tau_{j-1}, \tau_j; \Theta_j^*),$$

which would be available if the population coefficient matrices $\{\Theta_j^*\}_{j=1}^{k^*+1}$ were known. Under the conditions developed below, Step 1 consistently estimates the number of breaks and localizes them within neighborhoods that are asymptotically small relative to the adjacent seg-

ment lengths. Each preliminary segment is therefore dominated by observations from a single underlying regime, allowing $\{\widehat{\Theta}_j^r\}_{j=1}^{\hat{k}+1}$ to estimate their population counterparts sufficiently accurately. With these estimates held fixed, (11) approximates the oracle location optimization closely enough to deliver the refined estimators used in the limiting distribution theory.

The dynamic programming implementation of the two global optimization problems (8) and (11) is described in Section 2.4.

Step 2-2: Inference after global refinement. Accurate estimation of the break locations does not by itself quantify their uncertainty. Break-date inference is nonregular because both the localization scale and the limiting distribution depend on the magnitude of the structural change. Two asymptotic regimes therefore arise. For vanishing jumps, the rescaled localization error converges to the maximizer of a two-sided Brownian motion with linear drift. For non-vanishing jumps, the time discreteness remains asymptotically relevant and the unscaled error converges to the maximizer of a two-sided random walk. These two limits provide the basis for regime-specific confidence intervals for the refined break locations.

Specifically, recall $\widehat{\mathcal{T}}_k^r = (\hat{\tau}_1^r, \dots, \hat{\tau}_k^r)^\top, 0 = \hat{\tau}_0^r < \hat{\tau}_1^r < \dots < \hat{\tau}_k^r < \hat{\tau}_{k+1}^r = n$. On the event $\{\hat{k} = k^*\}$, the ordering of the estimated and true breaks provides the natural correspondence between $\hat{\tau}_j^r$ and τ_j^* for each $j \in [k^*]$. Define the nodewise jump at the j th break and its aggregate magnitude by $\Psi_j^{a*} = \theta_{j+1}^{a*} - \theta_j^{a*}, \kappa_j = \left\{ \frac{1}{p} \sum_{a=1}^p \|\Psi_j^{a*}\|_2^2 \right\}^{1/2}$. The results in Section 3.4 show that $|\hat{\tau}_j^r - \tau_j^*| = O_p(\kappa_j^{-2})$ in the vanishing-jump regime $\kappa_j \rightarrow 0$, whereas $|\hat{\tau}_j^r - \tau_j^*| = O_p(1)$ in the non-vanishing-jump regime $\kappa_j \rightarrow \varrho_j \in (0, \infty)$.

The limiting distributions involve population nuisance quantities whose definitions and regularity conditions are given in Assumption 5 and Section 3.4. Let $\mathbb{B}_{1,j}$ and $\mathbb{B}_{2,j}$ be independent standard Brownian motions on $[0, \infty)$. For the vanishing- and non-vanishing-jump regimes, respectively, define

$$\mathbb{Z}_j^v(r) = \begin{cases} 2\sigma_{1,j}\mathbb{B}_{1,j}(-r) - \varpi_{1,j}|r|, & r < 0, \\ 0, & r = 0, \\ 2\sigma_{2,j}\mathbb{B}_{2,j}(r) - \varpi_{2,j}|r|, & r > 0, \end{cases} \text{ and } \mathbb{Z}_j^{\text{nv}}(m) = \begin{cases} \sum_{\ell=1}^{-m} \xi_{1,j,\ell}, & m < 0, \\ 0, & m = 0, \\ \sum_{\ell=1}^m \xi_{2,j,\ell}, & m > 0, \end{cases} \quad m \in \mathbb{Z}. \quad (12)$$

The limiting increments $\{\xi_{q,j,\ell}\}_{\ell \geq 1}$ are specified in Assumption 5(c). Throughout, $\arg \max$ denotes the smallest maximizer when the maximizer is not unique.

Let $q_{j,\alpha/2}^v$ and $q_{j,1-\alpha/2}^v$ denote the corresponding quantiles of $\arg \max_{r \in \mathbb{R}} \mathbb{Z}_j^v(r)$, and let $q_{j,\alpha/2}^{\text{nv}}$ and $q_{j,1-\alpha/2}^{\text{nv}}$ denote the corresponding quantiles of $\arg \max_{m \in \mathbb{Z}} \mathbb{Z}_j^{\text{nv}}(m)$. The oracle marginal $(1 -$

α) confidence intervals for τ_j^* in the vanishing- and non-vanishing-jump regimes are, respectively,

$$\text{CI}_{j,1-\alpha}^{\text{v}} = \left[\hat{\tau}_j^r - \frac{q_{j,1-\alpha/2}^{\text{v}}}{\kappa_j^2}, \hat{\tau}_j^r - \frac{q_{j,\alpha/2}^{\text{v}}}{\kappa_j^2} \right] \text{ and } \text{CI}_{j,1-\alpha}^{\text{nv}} = \left[\hat{\tau}_j^r - q_{j,1-\alpha/2}^{\text{nv}}, \hat{\tau}_j^r - q_{j,\alpha/2}^{\text{nv}} \right]. \quad (13)$$

The limiting distributions and asymptotic coverage guarantees for the oracle intervals are established formally in Section 3.4. Because the oracle intervals depend on unknown population nuisance quantities, Section ?? of the Appendix develops their estimation and the corresponding feasible confidence intervals. Their finite-sample coverage properties are examined in the simulation studies.

Algorithm 1 GRIP-NET($\mathbf{X}_{[n]}, \mathcal{T}_{\tilde{k}}^g, \lambda^g, \lambda^r, \gamma$)

Require: Data $\mathbf{X}_{[n]}$, working locations $\mathcal{T}_{\tilde{k}}^g$, tuning parameters $\lambda^g, \lambda^r, \gamma$.

Ensure: Preliminary locations $\hat{\mathcal{T}}_{\tilde{k}}$ and refined locations $\hat{\mathcal{T}}_{\tilde{k}}^r$.

- 1: Estimate $\hat{\Theta}_{[\tilde{k}+1]}^g$ on the intervals induced by $\mathcal{T}_{\tilde{k}}^g$ using (4).
 - 2: Solve (8) by Algorithm 2 and set $\hat{\mathcal{T}}_{\tilde{k}} = \text{Uni}(\hat{\mathcal{U}}_{\tilde{k}})$.
 - 3: Estimate $\hat{\Theta}_{[\tilde{k}+1]}^r$ on the intervals induced by $\hat{\mathcal{T}}_{\tilde{k}}$ using (4).
 - 4: Solve (11) using the same dynamic programming principle as in the preliminary search.
-

2.4 Dynamic Programming Implementation

Because the provisional coefficient matrices are held fixed in Step 1, the relaxed global optimization in (8) admits a standard Bellman recursion. For $k \in \{0, \dots, \tilde{k}\}$ and $v \in [n]$, define

$$\mathcal{F}(v; \hat{\Theta}_{[\tilde{k}+1]}^g) = \min_{u \in \mathbb{T}_k(v)} \left\{ \sum_{j=1}^{k+1} \mathcal{L}(u_{j-1}, u_j; \hat{\Theta}_j^g) + \gamma \sum_{j=1}^{k+1} \mathbf{1}_{\{u_{j-1} < u_j\}} \right\}. \quad (14)$$

The terminal value $\mathcal{F}(n; \hat{\Theta}_{[\tilde{k}+1]}^g)$ is therefore the minimum objective value in (8). Separating the rightmost interval gives the recursion

$$\begin{aligned} \mathcal{F}(v; \hat{\Theta}_1^g) &= \mathcal{L}(0, v; \hat{\Theta}_1^g) + \gamma \mathbf{1}_{\{v > 0\}}, \\ \mathcal{F}(v; \hat{\Theta}_{[k+1]}^g) &= \min_{0 \leq u \leq v} \left\{ \mathcal{F}(u; \hat{\Theta}_{[k]}^g) + \mathcal{L}(u, v; \hat{\Theta}_{k+1}^g) + \gamma \mathbf{1}_{\{u < v\}} \right\}, \quad k \in [\tilde{k}]. \end{aligned} \quad (15)$$

After the forward recursion, the minimizing locations are recovered by backtracking. Setting $\hat{u}_{\tilde{k}+1} = n$, for $j = \tilde{k}, \dots, 1$ we obtain

$$\hat{u}_j = \arg \min_{0 \leq u \leq \hat{u}_{j+1}} \left\{ \mathcal{F}(u; \hat{\Theta}_{[j]}^g) + \mathcal{L}(u, \hat{u}_{j+1}; \hat{\Theta}_{j+1}^g) + \gamma \mathbf{1}_{\{u < \hat{u}_{j+1}\}} \right\}. \quad (16)$$

Algorithm 2 summarizes the dynamic-programming implementation of the preliminary global

reassignment.

Algorithm 2 GRIP-NET Preliminary Search ($\mathbf{X}_{[n]}, \hat{\Theta}_{[\tilde{k}+1]}^g, \gamma$)

Require: Data $\mathbf{X}_{[n]}$, provisional coefficient matrices $\hat{\Theta}_{[\tilde{k}+1]}^g$, and $\gamma > 0$.

Ensure: Estimated number $\hat{k}$ and preliminary locations $\hat{\mathcal{T}}_{\hat{k}}$.

- 1: Compute the Bellman values in (15) for $k = 1, \dots, \tilde{k}$ and $v = 1, \dots, n$.
 - 2: Store one minimizing predecessor for each pair (k, v) .
 - 3: Recover $\hat{\mathcal{U}}_{\tilde{k}}$ by the backward recursion in (16).
 - 4: Set $\hat{\mathcal{T}}_{\hat{k}} \leftarrow \text{Uni}(\hat{\mathcal{U}}_{\tilde{k}})$ and $\hat{k} \leftarrow |\hat{\mathcal{T}}_{\hat{k}}|$.
-

The Step 2 refinement problem (11) is implemented by the same dynamic-programming principle. The main differences are that the number of retained breaks $\hat{k}$ is fixed, the optimization is over the strict space $\mathbb{T}_{\hat{k}}(n)$, and no model-selection penalty is required. Because the refitted matrices $\hat{\Theta}_{[\hat{k}+1]}^r$ remain fixed throughout this search, Step 2 also involves only loss evaluations and location optimization, with no additional network estimation for candidate segments.

Computational complexity and advantage. The computational gain of GRIP-NET comes from separating high-dimensional precision-network estimation from the two global location searches. Let $\mathbf{Graph}(n, p)$ denote the cost of estimating one nodewise coefficient matrix, corresponding to p high-dimensional Lasso problems. Conventional exact global segmentation requires such an estimate on $O(n^2)$ candidate intervals and therefore incurs graph-estimation cost $O\{n^2 \mathbf{Graph}(n, p)\}$. By contrast, GRIP-NET performs only $\tilde{k} + 1$ provisional fits in Step 1 and $\hat{k} + 1$ refits in Step 2. Conditional on these fitted matrices, the Bellman recursion in Step 1 evaluates $O(\tilde{k}n^2)$ candidate transitions, while the refinement recursion requires $O(\hat{k}n^2)$ operations. The resulting overall complexity is therefore $O\{(\tilde{k} + \hat{k} + 2)\mathbf{Graph}(n, p) + (\tilde{k} + \hat{k})n^2\}$. The two terms have fundamentally different computational meanings. The first corresponds to expensive high-dimensional network estimation, whereas the second consists only of fixed-parameter loss evaluations and dynamic-programming operations. Thus, GRIP-NET does not replace global optimization by a restricted or greedy search; instead, it retains the two global location optimizations while reducing the number of high-dimensional network fits from $O(n^2)$ to $O(\tilde{k} + \hat{k})$. When network fitting is the dominant operation and $\tilde{k}, \hat{k} \ll n$, as is typical in the structural-break settings considered here, this decoupling yields substantial computational savings.

3 Theoretical Guarantees

This section develops the theoretical guarantees for the proposed GRIP-GGM procedure. Sections 3.1 and 3.2 introduce the data-generating framework and some basic assumptions. Section

3.3 establishes the consistency and preliminary localization rate of the Step 1. Finally, Section 3.4 studies the refined estimators obtained in Step 2, deriving their improved localization rates, limiting distributions, and the resulting theoretical guarantees for change-point inference.

3.1 Basic definitions

We first specify how the time-varying graphical observations are generated under functional dependence. Let $\{\mathcal{X}_i\}_{i \in \mathbb{Z}}$ be i.i.d. random elements and define the innovation history $\mathcal{F}_i^{\mathcal{X}} = (\dots, \mathcal{X}_{i-1}, \mathcal{X}_i)$. For each regime $j \in [k^* + 1]$, let $G_{(j)} = (g_{(j),1}, \dots, g_{(j),p})^\top$ be a measurable map from the innovation history to $\mathbb{R}^p$.

Definition 1 (Piecewise stationary GGM process). *For each $j \in [k^* + 1]$, let $\{\mathbf{X}_{(j),i}\}_{i \in \mathbb{Z}}$ be the stationary regime-specific process defined by*

$$\mathbf{X}_{(j),i} = (X_{(j),i,1}, \dots, X_{(j),i,p})^\top = G_{(j)}(\mathcal{F}_i^{\mathcal{X}}), \quad i \in \mathbb{Z}. \quad (17)$$

The observed nonstationary process is obtained by joining these stationary processes at the true change points. Specifically, setting $\tau_0^ = -\infty$ and $\tau_{k^*+1}^* = +\infty$, define*

$$\mathbf{X}_i = \sum_{j=1}^{k^*+1} \mathbf{X}_{(j),i} \mathbf{1}_{\{\tau_{j-1}^* < i \leq \tau_j^*\}}, \quad i \in \mathbb{Z}. \quad (18)$$

Definition 1 allows the data-generating map to change across regimes. Each regime-specific process is stationary, whereas the observed process is generally nonstationary because $G_{(j)}$ changes at $\tau_1^*, \dots, \tau_{k^*}^*$. To quantify temporal dependence, we examine how the observed process responds to perturbations of its underlying innovations. This requires a coupled version of the process in which selected innovations are replaced by independent copies while all remaining innovations are held fixed.

Definition 2 (Coupled process). *Let $\{\mathcal{X}'_t\}_{t \in \mathbb{Z}}$ be an independent copy of $\{\mathcal{X}_t\}_{t \in \mathbb{Z}}$. For integers $t_2 \leq t_1$, let $\mathcal{F}_{i,\{t_1,t_2\}}^{\mathcal{X}}$ be obtained from $\mathcal{F}_i^{\mathcal{X}}$ by replacing $\mathcal{X}_t$ with $\mathcal{X}'_t$ for every $t_2 \leq t \leq t_1$. Innovations occurring after time i are understood to have no effect on $\mathcal{F}_i^{\mathcal{X}}$. Define*

$$\mathbf{X}_{(j),i,\{t_1,t_2\}} = G_{(j)}\left(\mathcal{F}_{i,\{t_1,t_2\}}^{\mathcal{X}}\right), \quad j \in [k^* + 1]. \quad (19)$$

For a single replacement time t , write $\mathcal{F}_{i,\{t\}}^{\mathcal{X}} = \mathcal{F}_{i,\{t,t\}}^{\mathcal{X}}$ and $\mathbf{X}_{(j),i,\{t\}} = \mathbf{X}_{(j),i,\{t,t\}}$. In particular, if $t > i$, then $\mathbf{X}_{(j),i,\{t\}} = \mathbf{X}_{(j),i}$. The corresponding coupled observed process is

$$\mathbf{X}_{i,\{t_1,t_2\}} = \sum_{j=1}^{k^*+1} \mathbf{X}_{(j),i,\{t_1,t_2\}} \mathbf{1}_{\{\tau_{j-1}^* < i \leq \tau_j^*\}}. \quad (20)$$

Definition 3 (Functional dependence measures). For $\mathbf{v} \in \mathbb{R}^p$ and $q > 0$ satisfying $\sup_{i \in \mathbb{Z}} \|\mathbf{v}^\top \mathbf{X}_i\|_{L^q} < \infty$, define the directional functional dependence measure by

$$\delta_{t,q}^{\mathbf{X}}(\mathbf{v}) = \sup_{i \in \mathbb{Z}} \left\| \mathbf{v}^\top \mathbf{X}_i - \mathbf{v}^\top \mathbf{X}_{i,\{i-t\}} \right\|_{L^q}, \quad t \geq 0. \quad (21)$$

The uniform functional dependence measure and its cumulative version are defined as

$$\delta_{t,q}^{\mathbf{X}} = \sup_{\|\mathbf{v}\|_2=1} \delta_{t,q}^{\mathbf{X}}(\mathbf{v}), \quad \Delta_{m,q}^{\mathbf{X}} = \sum_{t=m}^{\infty} \delta_{t,q}^{\mathbf{X}}, \quad m \geq 0. \quad (22)$$

The quantity $\delta_{t,q}^{\mathbf{X}}$ measures the effect of replacing an innovation t periods in the past, uniformly over all unit directions. Its cumulative version $\Delta_{m,q}^{\mathbf{X}}$ summarizes the persistence of the effects generated by innovations at least m lags in the past.

3.2 Model Assumptions

Based on the above definitions, we next introduce some model assumptions.

Assumption 1 (Functional dependence). The random vectors $\{\mathbf{X}_i\}_{i=1}^n$ are generated by the regime-switching mechanism in (18). For each $j \in [k^* + 1]$, the stationary process $\{\mathbf{X}_{(j),i}\}_{i \in \mathbb{Z}}$ satisfies $\mathbb{E}(\mathbf{X}_{(j),i}) = \mathbf{0}_p$ and $\text{Cov}(\mathbf{X}_{(j),i}) = \Sigma_j^*$. In addition, the following conditions hold.

- a. There exist constants $\zeta_1, c_0, C_{\mathbf{X}1} > 0$ such that $\sup_{m \geq 0} \exp(c_0 m^{\zeta_1}) \Delta_{m,4}^{\mathbf{X}} \leq C_{\mathbf{X}1}$.
- b. There exist constants $\zeta_2 \in (0, 2]$ and $C_{\mathbf{X}2} > 0$ such that, for every $\mathbf{v} \in \mathbb{R}^p$ with $\|\mathbf{v}\|_2 = 1$ and every $t > 0$, $\sup_{i \in \mathbb{Z}} \mathbb{P}\{|\mathbf{v}^\top \mathbf{X}_i| > t\} \leq 2 \exp\{-(t/C_{\mathbf{X}2})^{\zeta_2}\}$.

Define the combined dependence–tail parameter

$$\zeta = (\zeta_1^{-1} + 2\zeta_2^{-1})^{-1}. \quad (23)$$

Throughout the theory we work under $\zeta < 1$.

Assumption 1 provides the basic regularity conditions for the time-dependent GGM change-point model. Condition (a) controls temporal dependence through the cumulative functional dependence measure $\Delta_{m,4}^{\mathbf{X}}$ and requires it to decay at a stretched-exponential rate with the lag m . Condition (b) imposes a uniform sub-Weibull tail bound on all one-dimensional projections of the observed process. The parameter ζ_2 characterizes the marginal tail behavior, with $\zeta_2 = 1$ corresponding to sub-exponential tails and $\zeta_2 = 2$ to sub-Gaussian tails.

The combined parameter in (23) captures the joint effect of temporal dependence and marginal tail behavior on the concentration inequalities used below. The factor 2 in $2\zeta_2^{-1}$

arises because the nodewise GGM analysis involves quadratic and cross-product sequences, such as $\varepsilon_{i,a}\mathbf{X}_{i,-a}$ and $\mathbf{X}_{i,-a}\mathbf{X}_{i,-a}^\top$. Under the convention $\zeta_2 \in (0, 2]$, the restriction $\zeta < 1$ holds automatically for any $\zeta_1 > 0$; tails lighter than sub-Gaussian can be accommodated by setting $\zeta_2 = 2$. A larger value of ζ corresponds to faster decay of temporal dependence and/or lighter marginal tails, and therefore leads to stronger concentration and more favorable estimation and localization rates. Overall, this formulation allows the graphical structure to vary across regimes while permitting general temporal dependence and sub-Weibull marginal behavior within each stationary segment.

Assumption 2 (Regime-specific graphical structure). *We assume the following conditions.*

- a. For some positive constants $\underline{\rho}_x$ and $\bar{\rho}_x$, the population covariance matrices $\{\boldsymbol{\Sigma}_j^*\}_{j=1}^{k^*+1}$ satisfy $0 < \underline{\rho}_x \leq \Lambda_{\min}(\boldsymbol{\Sigma}_j^*) \leq \Lambda_{\max}(\boldsymbol{\Sigma}_j^*) \leq \bar{\rho}_x < \infty$, $\forall j \in [k^* + 1]$, yielding uniformly bounded condition numbers $C_\rho := \bar{\rho}_x / \underline{\rho}_x$.*
- b. For each $j \in [k^* + 1]$ and $a \in [p]$, there is a support set $S_j^a \subset [p] \setminus \{a\}$ with $|S_j^a| \leq s$, such that the b -th entry of $\boldsymbol{\theta}_j^{a*}$ is zero whenever $b \in (S_j^a)^c := [p] \setminus \{S_j^a \cup \{a\}\}$.*

Assumption 2 collects the structural conditions on the regime-specific graphical models. Part (a) imposes uniform well-conditioning of the covariance matrices, which ensures that the nodewise regression representation is stable across all stationary regimes. Part (b) is the graph sparsity condition: for each node a , the support of $\boldsymbol{\theta}_j^{a*}$ coincides with the neighborhood of node a in the j th precision matrix, and therefore each node is connected to at most s other nodes; see, for example, [Meinshausen and Bühlmann \(2006\)](#); [Yuan \(2010\)](#).

We next present the signal-to-noise ratio (SNR) condition required for successful change-point detection.

Assumption 3 (Signal-of-noise ratio). *Denote $\boldsymbol{\Psi}_j^{a*} = \boldsymbol{\theta}_{j+1}^{a*} - \boldsymbol{\theta}_j^{a*}$ for $j \in [k^*]$ and $a \in [p]$. Let the minimal jump size be $\kappa = \min_{j \in [k^*]} \kappa_j = \min_{j \in [k^*]} \left\{ \frac{1}{p} \sum_{a=1}^p \|\boldsymbol{\Psi}_j^{a*}\|_2^2 \right\}^{1/2}$. Define the minimal spacing as $\Delta = \min_{j \in [k^*+1]} (\tau_j^* - \tau_{j-1}^*)$ and $\Delta_n = \Delta/n$. Let $r_\zeta(n, p) = \{s \log(np)\}^{2/\zeta-1}$. There exists a sufficiently large absolute constant C_{snr} and a diverging sequence $\{\Upsilon_n\}_{n \in \mathbb{N}_+}$ such that*

- a. $\Delta \kappa^2 \geq C_{snr} \Upsilon_n r_\zeta(n, p)$.*
- b. In addition, we further assume that $\{\Upsilon_n\}_{n \in \mathbb{N}_+}$ diverges faster than $s\{\log(np)\}^{2/\zeta}$, i.e., $\Upsilon_n \gg s\{\log(np)\}^{2/\zeta}$.*

The product $\Delta \kappa^2$ is the effective signal-to-noise ratio for offline change-point localization (e.g., [Rinaldo et al., 2021](#); [Kaul et al., 2023](#)). Assumption 3(a) imposes the lower bound needed for consistent detection and localization. The factor $r_\zeta(n, p)$ records the cost of estimating sparse

nodewise regressions under temporal dependence and non-Gaussian tails. In the benchmark case of independent sub-Gaussian observations, corresponding to $\zeta_1 = \infty$, $\zeta_2 = 2$, and hence $\zeta = 1$, we have $r_\zeta(n, p) = s \log(np)$, recovering the usual high-dimensional scaling up to the diverging factor Υ_n . When temporal dependence is stronger or the tails are heavier, $\zeta < 1$ and the exponent $2/\zeta - 1 > 1$ increases, thereby inflating both the logarithmic factor and the effective sparsity cost.

The sequence Υ_n separates estimation from inference. Under Assumption 3(a), it drives the preliminary localization error below the segment length, so that $\max_{1 \leq j \leq \hat{k}} |\hat{\tau}_j - \tau_j^*| = o_p(\Delta)$. Assumption 3(b) is stronger and is imposed for the refined estimators used in the limiting distribution and confidence interval theory. In particular, it provides the uniform tightness needed after global refinement. In the independent sub-Gaussian special case $\zeta = 1$, part (b) reduces to $\Upsilon_n \gg s \log^2(np)$; for non-vanishing jumps this yields the familiar sufficient scaling $s^2 \log^3(np) = o(\Delta)$. For $\zeta < 1$, the same interpretation remains, but the admissible regimes are correspondingly more restrictive because temporal dependence and heavier tails weaken concentration.

We next state the condition on the guessed change-points used to construct preliminary graphical parameter estimators. Recall $\mathcal{T}_k^g = \{\tau_1^g, \dots, \tau_k^g\} \in \mathbb{T}_k(n)$ with $\tau_0^g = 0, \tau_{k+1}^g = n$.

Assumption 4 (Guessed change-points). *There exist constants $\iota \in (0, 1/2]$ and $C_d, C_\tau > 0$ such that the guessed change-points $\mathcal{T}_k^g$ satisfy the following conditions.*

- a. *The number of guessed change-points is larger than that of the true change-points $\tilde{k} \geq k^*$.*
- b. *All guessed change-points are well separated: $\min_{j \in [\tilde{k}+1]} (\tau_j^g - \tau_{j-1}^g) \geq d_{\min}$ and $d_{\min} \geq C_d \Delta / \Upsilon_n^{1-2\iota}$.*
- c. *Each true change-point has at least one guessed change-point in its fractional neighborhood. That is, for any constant $\tilde{\iota} \in (1 - 2\iota, 1 - \iota]$, there exists a marker set $\mathcal{M} \subseteq [\tilde{k}]$ and an order-preserving map $\mathbf{m} : [k^*] \rightarrow \mathcal{M}$ such that $\max_{j \in [k^*]} |\tau_{\mathbf{m}(j)}^g - \tau_j^*| \leq \tilde{\tau} := \frac{C_\tau \Delta \kappa}{\sqrt{s \Upsilon_n^{2\tilde{\iota}}}}$.*

Assumption 4(a) allows the guessed set to over-specify the number of change points, whereas part (b) prevents the induced guessed intervals from becoming too short for stable graphical estimation. Part (c) ensures that every true change point is represented by at least one nearby guessed location. The admissible neighborhood radius $\tilde{\tau} = \frac{C_\tau \Delta \kappa}{\sqrt{s \Upsilon_n^{2\tilde{\iota}}}}$ increases with the segment length and the jump magnitude. This is natural: when more observations are available or the graphical change is stronger, the guessed location may be less accurate while still yielding informative provisional graphical parameter estimates. At the same time, the factor $\Upsilon_n^{-\tilde{\iota}}$ ensures that the neighborhood remains small relative to the segment length. In particular, when $\kappa = O(1)$ and $s = O(1)$, we have $\frac{\tilde{\tau}}{\Delta} = O(\Upsilon_n^{-\tilde{\iota}}) = o(1)$ so that $\tilde{\tau} = o(\Delta)$. Hence, the assumption only

requires each true change point to have one guessed representative within a vanishing fraction of the adjacent segment length and it does not require the guessed locations themselves to attain the final localization rate.

Remark 3.1. *Assumption 4 does not require the initializers themselves to be accurate change-point estimators. Instead, it requires only sufficiently informative coarse coverage of the underlying regimes, so that the provisional graphical estimates capture the relevant regime-specific structures. In practice, the appropriate resolution is unknown. We therefore construct equally spaced initializers and select their number using the validation procedure described in Section 4.1.*

3.3 Preliminary Estimation and Localization

We are now ready to state the preliminary localization result for $\widehat{\mathcal{T}}_k$.

Theorem 3.2 (Preliminary localization). *Suppose Assumptions 1, 2, 3(a), and 4 hold. Let the guessed graphical estimators $\widehat{\Theta}_{[k+1]}^g$ in (7) be constructed from $\mathcal{T}_k^g$ with $\lambda^g = C_g \left\{ \log^{1/2}(np) + \frac{\bar{\tau}}{d_{\min}^{1/2}} \right\}$. Let $\widehat{\mathcal{T}}_k$ be obtained from the preliminary global reassignment problem (8), with the penalty satisfying $C_\gamma r_\zeta(n, p) \leq \gamma \leq c_\gamma \Delta \kappa^2$ for sufficiently large $C_\gamma > 0$ and sufficiently small $c_\gamma > 0$. Then, for an absolute constant $C_e > 0$, with probability at least $1 - o(1)$, we have*

$$\hat{k} = k^*, \quad \max_{1 \leq j \leq k^*} \kappa_j^2 |\hat{\tau}_j - \tau_j^*| \leq C_e r_\zeta(n, p) = C_e s \log(np)^{2/\zeta-1}.$$

Theorem 3.2 first establishes selection consistency for the number of change-points, namely, $\Pr(\hat{k} = k^*) \rightarrow 1$. Thus, although the guessed set may over-specify the number of structural changes, the preliminary global reassignment step asymptotically removes redundant candidates while retaining all genuine change-points. Second, the theorem yields the change-point-specific localization rate $|\hat{\tau}_j - \tau_j^*| = O_{\mathbb{P}} \left\{ \frac{r_\zeta(n, p)}{\kappa_j^2} \right\}, j \in [k^*]$. The localization accuracy is determined by the jump size κ_j and the dependence-tail complexity factor $r_\zeta(n, p) = \{s \log(np)\}^{2/\zeta-1}$ with $\zeta = (\zeta_1^{-1} + 2\zeta_2^{-1})^{-1}$. Here, ζ_1 characterizes the decay rate of temporal dependence, while ζ_2 is the sub-Weibull tail parameter. Larger values of ζ_1 and ζ_2 correspond, respectively, to weaker temporal dependence and lighter tails, leading to a smaller localization error. In the special case of independent sub-Gaussian observations, $\zeta_1 = \infty$ and $\zeta_2 = 2$, so that $\zeta = 1$ and $r_\zeta(n, p) = s \log(np)$.

3.4 Global Refinement and Limiting Distributions

We next state the limiting distributions of the refined estimators delivered by Step 2 of GRIP-GGM. By Theorem 3.2, $\mathbb{P}(\hat{k} = k^*) \rightarrow 1$. Hence, on the event $\{\hat{k} = k^*\}$, the refined estimators can be matched in increasing order with the true change-points, and we write them as $\{\hat{\tau}_j^r\}_{j=1}^{k^*}$. For

$j \in [k^*]$ and $a \in [p]$, recall the nodewise jump $\Psi_j^{a*} = \theta_{j+1}^{a*} - \theta_j^{a*}$, and $\kappa_j = \left\{ \frac{1}{p} \sum_{a=1}^p \|\Psi_j^{a*}\|_2^2 \right\}^{1/2}$. The following assumption records the nuisance limits needed for inference. It is imposed after the detection assumptions because it concerns the local distributional limit of the refined estimator rather than the consistency of the preliminary segmentation.

Assumption 5 (Inference regularity). *For each $j \in [k^*]$, suppose the following limits exist and are finite, with the drift constants strictly positive.*

- a. *In the vanishing regime $\kappa_j \rightarrow 0$, there exist $\varpi_{1,j}, \varpi_{2,j} > 0$ such that $\frac{1}{p\kappa_j^2} \sum_{a=1}^p (\Psi_j^{a*})^\top \Sigma_j^* \Psi_j^{a*} \rightarrow \varpi_{1,j}$, and $\frac{1}{p\kappa_j^2} \sum_{a=1}^p (\Psi_j^{a*})^\top \Sigma_{j+1}^* \Psi_j^{a*} \rightarrow \varpi_{2,j}$.*
- b. *In the vanishing regime, there exist $\sigma_{1,j}^2, \sigma_{2,j}^2 > 0$ such that,*

$$\frac{1}{\kappa_j^2} \text{Var} \left(\frac{1}{p(\tau_j^* - \tau_{j-1}^*)^{1/2}} \sum_{i=\tau_{j-1}^*+1}^{\tau_j^*} \sum_{a=1}^p \varepsilon_{i,a} \mathbf{X}_i^\top \Psi_j^{a*} \right) \rightarrow \sigma_{1,j}^2,$$

$$\frac{1}{\kappa_j^2} \text{Var} \left(\frac{1}{p(\tau_{j+1}^* - \tau_j^*)^{1/2}} \sum_{i=\tau_j^*+1}^{\tau_{j+1}^*} \sum_{a=1}^p \varepsilon_{i,a} \mathbf{X}_i^\top \Psi_j^{a*} \right) \rightarrow \sigma_{2,j}^2,$$

where $\varepsilon_{i,a}$ denotes the nodewise regression error for node a in the corresponding segment.

- c. *In the non-vanishing regime $\kappa_j \rightarrow \varrho_j \in (0, \infty)$, define the one-step local loss increments*

$$\Xi_{1,j,\ell}^{(n)} = \mathcal{L}(\tau_j^* - \ell, \tau_j^* - \ell + 1; \Theta_j^*) - \mathcal{L}(\tau_j^* - \ell, \tau_j^* - \ell + 1; \Theta_{j+1}^*), \quad \ell \geq 1,$$

and

$$\Xi_{2,j,\ell}^{(n)} = \mathcal{L}(\tau_j^* + \ell - 1, \tau_j^* + \ell; \Theta_{j+1}^*) - \mathcal{L}(\tau_j^* + \ell - 1, \tau_j^* + \ell; \Theta_j^*), \quad \ell \geq 1.$$

The finite-dimensional distributions of $\{\Xi_{q,j,\ell}^{(n)}\}_{\ell \geq 1}$ converge to those of $\{\xi_{q,j,\ell}\}_{\ell \geq 1}$ for $q = 1, 2$, where the two limiting sequences are mutually independent and have negative means and finite variances:

$$[\mathbb{E}(\xi_{q,j,\ell}) = -\varrho_j^2 \varpi_{q,j} < 0, \quad \text{Var}(\xi_{q,j,\ell}) = 4\varrho_j^2 \bar{\sigma}_{q,j}^2 < \infty.] \text{ (why this form???)}$$

The limiting processes $\mathbb{Z}_j^v$ and $\mathbb{Z}_j^{nv}$ are defined in (??) and (??) of Section 2.3. The nuisance constants in Assumption 5 are functionals of the two adjacent stationary regimes, the population nodewise coefficients Θ_j^* and Θ_{j+1}^* , and the temporal dependence structure in Assumption 1.

Theorem 3.3 (Limiting distributions of refined change-points). *Suppose the conditions of Theorem 3.2, Assumption 3(b), and Assumption 5 hold. Let the refitted graphical estimators $\hat{\Theta}_{[k+1]}^r$*

in (10) be constructed from the preliminary segmentation $\hat{\mathcal{T}}_{\hat{k}}$ with $\lambda^r = C_r \log^{1/2}(np)$, where $C_r > 0$ is a sufficiently large constant. Let $\{\hat{\tau}_j^r\}_{j=1}^{\hat{k}}$ be obtained from the Step 2 global refinement problem (11). Then, on the event $\{\hat{k} = k^*\}$, for every fixed $j \in [k^*]$, the following statements hold.

a. If $\kappa_j \rightarrow \varrho_j \in (0, \infty)$, then

$$|\hat{\tau}_j^r - \tau_j^*| = O_p(1), \quad \hat{\tau}_j^r - \tau_j^* \Rightarrow \arg \max_{m \in \mathbb{Z}} \mathbb{Z}_j^{\text{nv}}(m). \quad (24)$$

b. If $\kappa_j \rightarrow 0$, then

$$|\hat{\tau}_j^r - \tau_j^*| = O_p(\kappa_j^{-2}), \quad \kappa_j^2(\hat{\tau}_j^r - \tau_j^*) \Rightarrow \arg \max_{r \in \mathbb{R}} \mathbb{Z}_j^{\text{v}}(r). \quad (25)$$

Theorem 3.3 separates the two asymptotic regimes for graphical change-point inference. When the jump size does not vanish, the discreteness of the time index remains visible and the refined estimator converges to the maximizer of the two-sided random-walk process. When the jump size vanishes, the local contrast accumulates over a growing neighborhood of width κ_j^{-2} , and the limiting process is the two-sided Brownian process with negative linear drift introduced there.

The limiting distributions in Theorem 3.3 provide the quantiles required to construct confidence intervals for individual change-point locations. Consequently, the oracle intervals in (13) and (??) achieve asymptotically correct componentwise coverage.

Corollary 3.4 (Joint convergence of refined change-point estimators). *Suppose the conditions of Theorem 3.3 hold and that k^* is fixed. For each $j \in [k^*]$, define*

$$a_{j,n} = \begin{cases} 1, & \kappa_j \rightarrow \varrho_j \in (0, \infty), \\ \kappa_j^2, & \kappa_j \rightarrow 0, \end{cases} \quad \mathcal{Z}_j = \begin{cases} \arg \max_{m \in \mathbb{Z}} \mathbb{Z}_j^{\text{nv}}(m), & \kappa_j \rightarrow \varrho_j \in (0, \infty), \\ \arg \max_{r \in \mathbb{R}} \mathbb{Z}_j^{\text{v}}(r), & \kappa_j \rightarrow 0. \end{cases}$$

Then, on the event $\{\hat{k} = k^*\}$,

$$(a_{1,n}(\hat{\tau}_1^r - \tau_1^*), \dots, a_{k^*,n}(\hat{\tau}_{k^*}^r - \tau_{k^*}^*)) \Rightarrow (\mathcal{Z}_1, \dots, \mathcal{Z}_{k^*}),$$

where $\mathcal{Z}_1, \dots, \mathcal{Z}_{k^*}$ are mutually independent. Consequently, the joint limiting distribution provides the calibration required for simultaneous confidence intervals for the multiple break locations.

The asymptotic independence in Corollary 3.4 follows from the global-to-local structure of the

refined estimator. Once the refined locations are restricted to their asymptotically relevant neighborhoods, the global criterion reduces to separate local contrast problems around the individual true breaks. Because distinct break points are separated by diverging segment lengths, these local neighborhoods become asymptotically disjoint under the functional-dependence framework.

4 Numerical studies

4.1 Data-adaptive tuning parameter selection

The implementation of GRIP-GGM in Algorithm 1 requires the tuning parameters λ^g , γ , and λ^r , together with an ordered collection of $\tilde{k}$ initial change points $\mathcal{T}_{\tilde{k}}^g$. Although the theoretical results are established for a fixed initializer set satisfying Assumption 4, an appropriate value of $\tilde{k}$ is generally unknown in applications. We therefore select the tuning parameters for each fixed initializer and then choose $\tilde{k}$ by comparing the corresponding validation scores.

Suppose that n is even and let $n_0 = n/2$. We construct two interlaced subsamples, $\mathbf{X}_i^o = \mathbf{X}_{2i-1}$ and $\mathbf{X}_i^e = \mathbf{X}_{2i}$, for $i = 1, \dots, n_0$, where o and e denote the odd and even observations, respectively. Let Λ_g , Γ , and Λ_r be finite candidate sets for λ^g , γ , and λ^r , respectively. We first describe the selection procedure for a fixed initializer count $\tilde{k}$.

Step 1: Selection for a fixed initializer. For a given $\tilde{k}$, define the equally spaced initializer

$$\mathcal{T}_{\tilde{k}}^{g,o} = \left\{ \left\lfloor \frac{jn_0}{\tilde{k}+1} \right\rfloor : j = 1, \dots, \tilde{k} \right\}. \quad (26)$$

Note that in the absence of prior information about the change-point locations, the equally spaced construction provides a simple and neutral way to distribute the initializers over the entire observation period. If $\tilde{k}$ is too small, some working intervals may span multiple regimes and obscure structural changes. If it is too large, the resulting short intervals may lead to unstable graphical estimates. This motivates the following cross validation procedure.

For each $(\lambda^g, \gamma, \lambda^r) \in \Lambda_g \times \Gamma \times \Lambda_r$, we apply GRIP-GGM in Algorithm 1 to the odd subsample $\mathbf{X}_i^o$ using $\mathcal{T}_{\tilde{k}}^{g,o}$. Denote the resulting change-point estimator by $\hat{\mathcal{T}}_{\tilde{k}^o}^o(\lambda^g, \gamma, \lambda^r; \tilde{k}) = \{\hat{\tau}_1^o, \dots, \hat{\tau}_{\tilde{k}^o}^o\}$ where $\tilde{k}^o$ is the corresponding estimated number of change points. Set $\hat{\tau}_0^o = 0$ and $\hat{\tau}_{\tilde{k}^o+1}^o = n_0$. Using these estimated change points, we estimate the graphical coefficients on each odd-subsample segment with λ^r :

$$\hat{\Theta}_j^o = \hat{\Theta}_{(\hat{\tau}_{j-1}^o, \hat{\tau}_j^o]}^{o, \lambda^r}, \quad j = 1, \dots, \tilde{k}^o + 1. \quad (27)$$

For an interval $(u, v]$ and a graphical coefficient matrix $\Theta = (\theta^1, \dots, \theta^p)^\top$, define the loss on the even subsample by $\mathcal{L}^e(u, v; \Theta) = \frac{1}{p} \sum_{a=1}^p \sum_{i=u+1}^v \left\{ X_{i,a}^e - (\mathbf{X}_i^e)^\top \theta^a \right\}^2$. The validation score is

then defined by

$$\mathcal{V}(\lambda^g, \gamma, \lambda^r; \tilde{k}) = \sum_{j=1}^{\hat{k}^\circ+1} \mathcal{L}^e \left(\hat{\tau}_{j-1}^\circ, \hat{\tau}_j^\circ; \hat{\Theta}_j^\circ \right). \quad (28)$$

Thus, the change-point locations and graphical coefficients are estimated from the odd observations, while their fit is evaluated using the corresponding even observations.

For the fixed initializer count $\tilde{k} \geq 1$, we select the tuning parameters by

$$\left(\lambda^{g,\diamond}(\tilde{k}), \gamma^\diamond(\tilde{k}), \lambda^{r,\diamond}(\tilde{k}) \right) \in \arg \min_{\lambda^g \in \Lambda_g, \gamma \in \Gamma, \lambda^r \in \Lambda_r} \mathcal{V}(\lambda^g, \gamma, \lambda^r; \tilde{k}). \quad (29)$$

The corresponding minimum validation score is defined as

$$\mathcal{V}^\diamond(\tilde{k}) = \mathcal{V} \left(\lambda^{g,\diamond}(\tilde{k}), \gamma^\diamond(\tilde{k}), \lambda^{r,\diamond}(\tilde{k}); \tilde{k} \right). \quad (30)$$

Step 2: Adaptive selection of the initializer. Note that the previous procedure associates each fixed initializer count $\tilde{k}$ with an optimally tuned parameter combination and the corresponding minimum validation score $\mathcal{V}^\diamond(\tilde{k})$. This provides a direct basis for selecting the initializer count. Specifically, we consider the candidate set $\mathcal{K} = \{1, \dots, k_{\max}\}$, where $k_{\max}$ is a prespecified upper bound. We repeat the fixed- $\tilde{k}$ tuning procedure for every $\tilde{k} \in \mathcal{K}$ and select the initializer count that achieves the smallest optimized validation score:

$$\tilde{k}^\diamond \in \arg \min_{\tilde{k} \in \mathcal{K}} \mathcal{V}^\diamond(\tilde{k}). \quad (31)$$

The final tuning parameters are those associated with the selected initializer count: $\lambda^{g,\diamond} = \lambda^{g,\diamond}(\tilde{k}^\diamond)$, $\gamma^\diamond = \gamma^\diamond(\tilde{k}^\diamond)$, $\lambda^{r,\diamond} = \lambda^{r,\diamond}(\tilde{k}^\diamond)$.

Together with the selected initializer count $\tilde{k}^\diamond$, these tuning parameters are then supplied to Algorithm 1. This yields the final estimated number of change points $\hat{k}$ and the corresponding change-point estimator $\hat{\mathcal{T}}_{\hat{k}}$. Algorithm 3 summarizes the complete procedure.

4.2 Model setups

We generate observations from model 1 with segment-specific precision matrices $\{\mathbf{\Omega}_j^*\}_{j=1}^{K^*+1}$. Adjacent segments alternate between two sparse positive-definite precision matrices, $\mathbf{\Omega}^{(1)}$ and $\mathbf{\Omega}^{(2)}$, with $\mathbf{\Omega}_j^* = \mathbf{\Omega}^{(1)}$ for odd j and $\mathbf{\Omega}_j^* = \mathbf{\Omega}^{(2)}$ for even j . **We consider three representative precision-network structures: Erdős-Rényi, hub, and chain networks.** For each structure, $\mathbf{\Omega}^{(1)}$ and $\mathbf{\Omega}^{(2)}$ share the same graph type but differ over a prescribed subset of nodes. For the hub and chain structures, “Node” denotes the number of affected nodes and therefore controls the overall magnitude of the structural change. The main comparison uses Node = 20.

Algorithm 3 Data-adaptive tuning for GRIP-GGM

Require: Data $\{\mathbf{X}_i\}_{i=1}^n$, upper bound $k_{\max}$, and candidate sets Λ_g , Γ , and Λ_r .

Ensure: Selected initializer count $\tilde{k}^\diamond$ and tuning parameters $\lambda^{g,\diamond}$, $\gamma^\diamond$, and $\lambda^{r,\diamond}$.

- 1: Construct the odd and even subsamples $\mathbf{X}_i^o$ and $\mathbf{X}_i^e$.
 - 2: **for** $\tilde{k} = 1, \dots, k_{\max}$ **do**
 - 3: Construct $\mathcal{T}_{\tilde{k}}^{g,o}$ according to (26).
 - 4: **for** each $(\lambda^g, \gamma, \lambda^r) \in \Lambda_g \times \Gamma \times \Lambda_r$ **do**
 - 5: Apply GRIP-GGM to the odd subsample using $\mathcal{T}_{\tilde{k}}^{g,o}$, λ^g , γ , and λ^r .
 - 6: Obtain the estimated number $\hat{k}^o$ and change-point set $\hat{\mathcal{T}}_{\hat{k}^o}^o$.
 - 7: Estimate $\{\hat{\Theta}_j^o\}_{j=1}^{\hat{k}^o+1}$ on the estimated odd-subsample segments using λ^r .
 - 8: Compute $\mathcal{V}(\lambda^g, \gamma, \lambda^r; \tilde{k})$ from the even subsample using (28).
 - 9: **end for**
 - 10: Select $\lambda^{g,\diamond}(\tilde{k})$, $\gamma^\diamond(\tilde{k})$, and $\lambda^{r,\diamond}(\tilde{k})$ according to (29).
 - 11: Record $\mathcal{V}^\diamond(\tilde{k})$.
 - 12: **end for**
 - 13: Select $\tilde{k}^\diamond$ according to (31).
 - 14: Set $\lambda^{g,\diamond} = \lambda^{g,\diamond}(\tilde{k}^\diamond)$, $\gamma^\diamond = \gamma^\diamond(\tilde{k}^\diamond)$, and $\lambda^{r,\diamond} = \lambda^{r,\diamond}(\tilde{k}^\diamond)$.
 - 15: **Return:** $\tilde{k}^\diamond$, $\lambda^{g,\diamond}$, $\gamma^\diamond$, and $\lambda^{r,\diamond}$.
-

To evaluate robustness to temporal dependence, each network configuration is considered under both IID and autoregressive observations. Under the IID setting, observations in segment j are independently generated as $\mathbf{X}_i \sim \mathcal{N}\{\mathbf{0}, (\mathbf{\Omega}_j^*)^{-1}\}$. Under the AR setting,

$$\mathbf{X}_i = 0.3\mathbf{X}_{i-1} + \sqrt{1 - 0.3^2} \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\{\mathbf{0}, (\mathbf{\Omega}_j^*)^{-1}\}, \quad \tau_{j-1}^* < i \leq \tau_j^*.$$

The scaling of the innovation term ensures that $(\mathbf{\Omega}_j^*)^{-1}$ is the stationary covariance associated with segment j , while the autoregressive recursion introduces serial dependence.

We consider two principal multiple-break configurations. Setting **C1** has $n = 400$, $p = 100$, $K^* = 2$, and break locations $(\tau_1^*, \tau_2^*) = (0.25n, 0.60n)$. Setting **C2** has $n = 1000$, $p = 100$, $K^* = 4$, and break locations $(\tau_1^*, \tau_2^*, \tau_3^*, \tau_4^*) = (0.15n, 0.40n, 0.60n, 0.80n)$. We compare GRIP with `changepointshd`, `LinearDetect`, and optimistic binary segmentation (OBS). The first two competitors are based on [Bybee and Atchadé \(2018\)](#) and [Bai and Safikhani \(2023b\)](#), respectively, while OBS follows [Kovács et al. \(2024b\)](#) and is implemented using the `hdcd` package. For the competing methods, tuning parameters are chosen following the recommendations in the corresponding papers or, when available, the default settings in their software implementations. All methods are evaluated using their final estimated change-point sets. As for our method, we use the data-adaptive procedure proposed in Section 4.1 to choose the tuning parameters and yield the final change point estimators. Results are based on 100 Monte Carlo replications.

In addition to the main comparison, we conduct a signal-strength experiment designed to examine the transition from preliminary detection to refined localization. Under the chain-network model and configuration **C1**, we vary `Node` $\in \{10, 12, 14, 16, 18\}$, so that the structural

signal increases with the number of affected nodes. This experiment compares the adaptive working configuration, the preliminary estimator, and the globally refined estimator, thereby isolating the empirical gain attributable to the refinement step.

Let $\mathcal{T}^* = \{\tau_1^*, \dots, \tau_{K^*}^*\}$ and $\hat{\mathcal{T}} = \{\hat{\tau}_1, \dots, \hat{\tau}_{\hat{K}}\}$ denote the true and estimated change-point sets. Estimation performance is measured by $E(\hat{K})$, $P(\hat{K} = K^*)$, and the scaled Hausdorff distance $d_H(\hat{\mathcal{T}}, \mathcal{T}^*)/n$. For inference, we report the marginal and simultaneous coverage probabilities of nominal 95% confidence intervals, together with their scaled lengths $(U_j - L_j)/n$. Taken together, these experiments assess change-point selection and localization, the additional statistical gain from global refinement, the finite-sample validity of break-date inference, and robustness to temporal dependence.

4.3 Change-Point Estimation

We first evaluate the ability of GRIP to recover the number and locations of multiple structural breaks. Figure 2 compares the distributions of the scaled Hausdorff distance for GRIP, `changepointsHD` (Bybee and Atchadé, 2018), and optimistic binary segmentation (OBS) (Kovács et al., 2024b) under configurations **C1** and **C2**. Tables 1 and 2 provide the corresponding numerical results for the estimated number of change points, $E(\hat{K})$, the selection probability $P(\hat{K} = K^*)$, and the mean and standard deviation of the scaled Hausdorff distance.

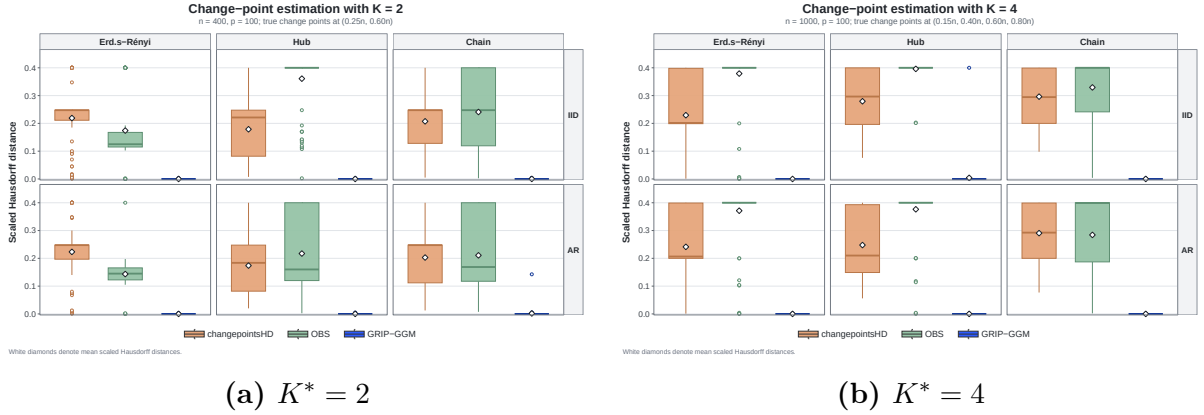


Figure 2: Boxplots of the scaled Hausdorff distance under the two- and four-change-point configurations. Panels (a) and (b) correspond to settings **C1** and **C2**, respectively. Results are shown for the three precision-network structures under IID and autoregressive observations.

Table 1: Change-point estimation performance under setting **C1**, with $K^* = 2$, $n = 400$, $p = 100$, and true change-point locations $(0.25n, 0.60n)$. Means and standard deviations are based on 100 replications.

Network	Dep.	Method	$E(\hat{K})$	$P(\hat{K} = K^*)$	Scaled HD	
					Mean	SD
Erdős–Rényi	IID	GRIP	2.00	1.00	0.0001	0.0004
Erdős–Rényi	IID	changepointsHD	1.09	0.21	0.2191	0.1114
Erdős–Rényi	IID	OBS	3.08	0.05	0.1741	0.1150
Erdős–Rényi	AR	GRIP	2.00	1.00	0.0001	0.0004
Erdős–Rényi	AR	changepointsHD	1.08	0.19	0.2233	0.1088
Erdős–Rényi	AR	OBS	4.28	0.04	0.1434	0.0460
Hub, Node = 20	IID	GRIP	2.00	1.00	0.0001	0.0004
Hub, Node = 20	IID	changepointsHD	1.48	0.50	0.1787	0.0965
Hub, Node = 20	IID	OBS	0.51	0.01	0.3610	0.0955
Hub, Node = 20	AR	GRIP	2.00	1.00	0.0001	0.0007
Hub, Node = 20	AR	changepointsHD	1.53	0.58	0.1740	0.0990
Hub, Node = 20	AR	OBS	2.91	0.01	0.2172	0.1231
Chain, Node = 20	IID	GRIP	2.00	1.00	0.0002	0.0007
Chain, Node = 20	IID	changepointsHD	1.33	0.38	0.2073	0.1029
Chain, Node = 20	IID	OBS	1.68	0.17	0.2409	0.1340
Chain, Node = 20	AR	GRIP	2.01	0.99	0.0017	0.0143
Chain, Node = 20	AR	changepointsHD	1.30	0.34	0.2029	0.0965
Chain, Node = 20	AR	OBS	2.34	0.14	0.2107	0.1246

Table 2: Change-point estimation performance under setting **C2**, with $K^* = 4$, $n = 1000$, $p = 100$, and true change-point locations $(0.15n, 0.40n, 0.60n, 0.80n)$. Results are based on 100 replications.

Network	Dep.	Method	$E(\hat{K})$	$P(\hat{K} = K^*)$	Scaled HD	
					Mean	SD
Erdős–Rényi	IID	GRIP	4.00	1.00	0.00000	0.00000
Erdős–Rényi	IID	changepointsHD	2.30	0.19	0.22637	0.12580
Erdős–Rényi	IID	OBS	0.23	0.04	0.37919	0.08488
Erdős–Rényi	AR	GRIP	4.00	1.00	0.00004	0.00020
Erdős–Rényi	AR	changepointsHD	2.02	0.14	0.25792	0.13080
Erdős–Rényi	AR	OBS	0.37	0.04	0.37134	0.09463
Hub, Node = 20	IID	GRIP	3.96	0.99	0.00403	0.04000
Hub, Node = 20	IID	changepointsHD	1.95	0.04	0.28493	0.11225
Hub, Node = 20	IID	OBS	0.04	0.00	0.39604	0.02786
Hub, Node = 20	AR	GRIP	4.00	1.00	0.00005	0.00022
Hub, Node = 20	AR	changepointsHD	2.20	0.08	0.25977	0.11573
Hub, Node = 20	AR	OBS	0.29	0.03	0.37639	0.08411
Chain, Node = 20	IID	GRIP	4.00	1.00	0.00011	0.00031
Chain, Node = 20	IID	changepointsHD	1.82	0.07	0.28794	0.10552
Chain, Node = 20	IID	OBS	1.03	0.06	0.32938	0.11666
Chain, Node = 20	AR	GRIP	4.00	1.00	0.00006	0.00024
Chain, Node = 20	AR	changepointsHD	1.78	0.04	0.27141	0.10013
Chain, Node = 20	AR	OBS	1.78	0.10	0.28425	0.13732

The results show a clear separation in both model selection and localization accuracy. Across all six network–dependence configurations in **C1**, GRIP selects the correct number of change points in at least 99% of the replications and produces very small scaled Hausdorff distances. The same pattern persists in the more complex four-break configuration **C2**: the correct number of breaks is selected with probability at least 0.99, and the mean scaled Hausdorff distance remains below 0.005 in every setting. The boxplots in Figure 2 further show that the localization errors are tightly concentrated near zero under both IID and autoregressive observations.

The competing procedures exhibit substantially larger localization errors and greater sensitivity to the underlying network structure. **changepointsHD** tends to underestimate the number of breaks, particularly under **C2**. **OBS** exhibits both under- and over-segmentation across the different network structures and dependence settings. Consequently, the final estimated change-point sets from these methods have markedly larger Hausdorff distances than those obtained by

GRIP.

4.4 Change-Point Inference

We next evaluate the finite-sample performance of the proposed break-date inference procedure based on the refined change-point estimators. The feasible 95% confidence intervals are constructed using the limiting distributions established in Section 3, with the required nuisance quantities estimated as described in the Supplementary Material. We report marginal coverage for each change point, simultaneous coverage for all change points, and scaled interval lengths.

Table 3: Finite-sample performance of 95% confidence intervals under setting **C1**, with $K^* = 2$, $n = 400$, $p = 100$, and break locations $(0.25n, 0.60n)$. Results are based on 100 replications.

Network	Dep.	Coverage probability			Scaled interval length		
		CP 1	CP 2	Simultaneous	CP 1	CP 2	Average
Erdős–Rényi	IID	0.99	1.00	0.99	0.323	0.375	0.349
Erdős–Rényi	AR	1.00	0.99	0.99	0.253	0.302	0.278
Hub, Node = 20	IID	1.00	1.00	1.00	0.118	0.142	0.130
Hub, Node = 20	AR	0.97	0.98	0.96	0.053	0.072	0.062
Chain, Node = 20	IID	0.97	0.95	0.92	0.107	0.121	0.114
Chain, Node = 20	AR	0.95	0.94	0.90	0.091	0.112	0.102

Table 4: Finite-sample performance of 95% confidence intervals under setting **C2**, with $K^* = 4$, $n = 1000$, $p = 100$, and break locations $(0.15n, 0.40n, 0.60n, 0.80n)$. Results are based on 100 replications.

Network	Dep.	Coverage probability					Scaled interval length				
		CP 1	CP 2	CP 3	CP 4	Simultaneous	CP 1	CP 2	CP 3	CP 4	Average
Erdős–Rényi	IID	1.00	1.00	1.00	1.00	1.00	0.376	0.424	0.416	0.402	0.405
Erdős–Rényi	AR	0.98	1.00	0.99	0.99	0.96	0.331	0.359	0.345	0.339	0.344
Hub, Node = 20	IID	0.97	0.99	0.98	0.99	0.96	0.095	0.095	0.098	0.098	0.096
Hub, Node = 20	AR	0.99	1.00	0.99	0.97	0.95	0.086	0.101	0.076	0.087	0.087
Chain, Node = 20	IID	0.97	0.99	0.97	0.96	0.89	0.125	0.149	0.147	0.149	0.142
Chain, Node = 20	AR	0.96	1.00	0.98	1.00	0.94	0.121	0.137	0.120	0.127	0.126

Tables 3 and 4 show that the marginal coverage probabilities are generally close to, and in several settings slightly above, the nominal 95% level under both IID and autoregressive observations. Simultaneous coverage is naturally more demanding and ranges from 0.90 to

1.00 under **C1** and from 0.89 to 1.00 under **C2**. The mild undercoverage in the chain-network setting is accompanied by substantially shorter intervals than those obtained for the Erdős–Rényi network, whereas the latter exhibits somewhat conservative coverage. Overall, the results provide finite-sample support for both componentwise and simultaneous break-date inference under multiple structural changes and temporal dependence.

Figures 3 and 4 provide a complementary visualization of the confidence intervals for the Erdős–Rényi and chain networks. Within each figure, rows correspond to the IID and AR settings and columns to the individual change points. Each interval is recentered so that zero represents the true break location. For visual clarity, only intervals containing the true location are displayed; the reported coverage probabilities are calculated using all 100 replications.

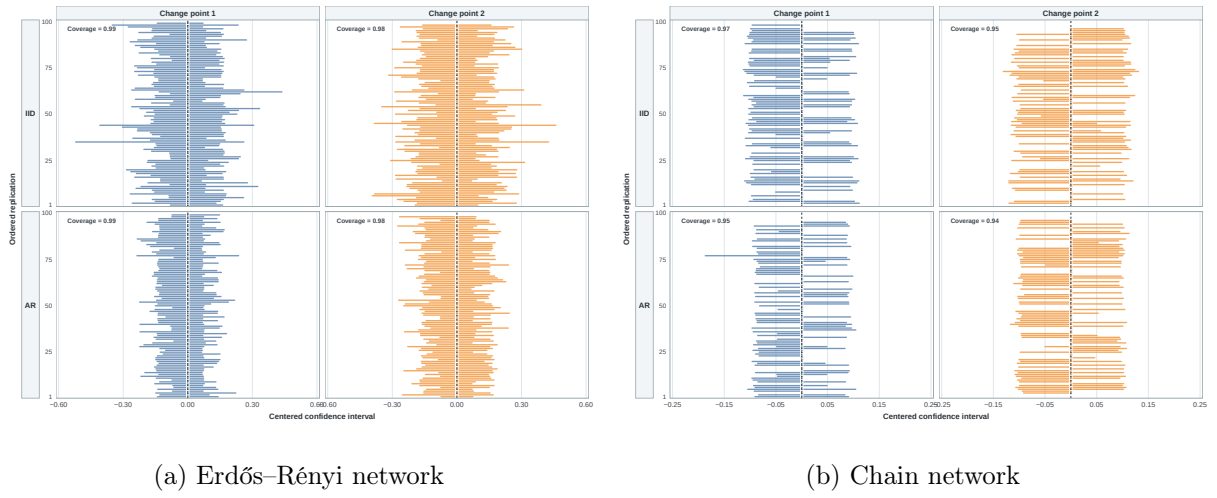


Figure 3: Empirical 95% confidence intervals under setting **C1**.

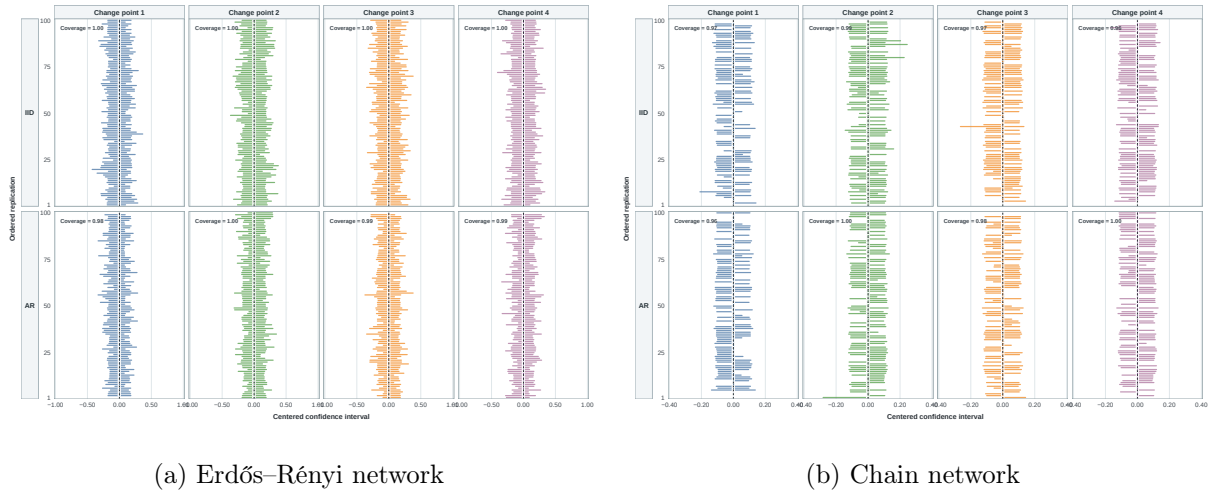


Figure 4: Empirical 95% confidence intervals under setting **C2**.

The visual results are consistent with the numerical coverage summaries: the refined break

estimators are concentrated around the true locations, and the inference procedure continues to perform similarly under IID and autoregressive observations.

4.5 Effect of Signal Strength and Improvement of Refinement

To examine the performance of the proposed procedure across different signal strengths, we consider the chain-network model with $n = 400$, $p = 100$, and two change points located at $(0.25n, 0.60n)$. The number of nodes affected by each structural change, denoted by “Node”, is varied from 10 to 18. A larger number of affected nodes corresponds to a stronger overall change in the precision-network structure. For each configuration, we conduct 100 replications under both IID and autoregressive observations.

We compare localization accuracy at three stages of the procedure. Specifically, HD_{guess} , HD_{pre} , and HD_{ref} denote the mean scaled Hausdorff distances of the guessed estimator, the preliminary estimator obtained from the first global optimization, and the refined estimator obtained after segment-wise refitting and the second global optimization, respectively. To quantify the localization gain from refinement, define

$$\Delta_{\text{guess}} = 100 \times \frac{\text{HD}_{\text{guess}} - \text{HD}_{\text{ref}}}{\text{HD}_{\text{guess}}}, \quad \Delta_{\text{pre}} = 100 \times \frac{\text{HD}_{\text{pre}} - \text{HD}_{\text{ref}}}{\text{HD}_{\text{pre}}}.$$

Table 5 and Figure 5 show a clear improvement as the structural signal strengthens. The probability of correctly estimating $K^* = 2$ increases from 0.20 to 0.99 under IID observations and from 0.49 to 0.98 under AR observations. Meanwhile, HD_{ref} decreases from 0.2002 to 0.0026 and from 0.1298 to 0.0030, respectively.

Table 5: Estimation and refinement performance under the chain-network model with $K^* = 2$ across different signal strengths. Results are based on 100 replications.

Dep.	Node	$E(\hat{K})$	$P(\hat{K} = K^*)$	HD_{guess}	HD_{pre}	HD_{ref}	Δ_{guess}	Δ_{pre}
IID	10	1.20	0.20	0.2185	0.2036	0.2002	8.4%	1.6%
IID	12	1.46	0.46	0.1750	0.1424	0.1354	22.6%	4.9%
IID	14	1.83	0.83	0.1130	0.0494	0.0428	62.1%	13.2%
IID	16	1.96	0.96	0.0960	0.0149	0.0104	89.2%	30.5%
IID	18	1.99	0.99	0.0943	0.0059	0.0026	97.2%	55.1%
AR	10	1.49	0.49	0.1679	0.1404	0.1298	22.7%	7.6%
AR	12	1.79	0.79	0.1211	0.0655	0.0537	55.7%	18.0%
AR	14	1.93	0.93	0.0949	0.0293	0.0184	80.6%	37.2%
AR	16	1.98	0.96	0.0950	0.0164	0.0094	90.1%	42.4%
AR	18	2.02	0.98	0.0893	0.0060	0.0030	96.6%	50.0%

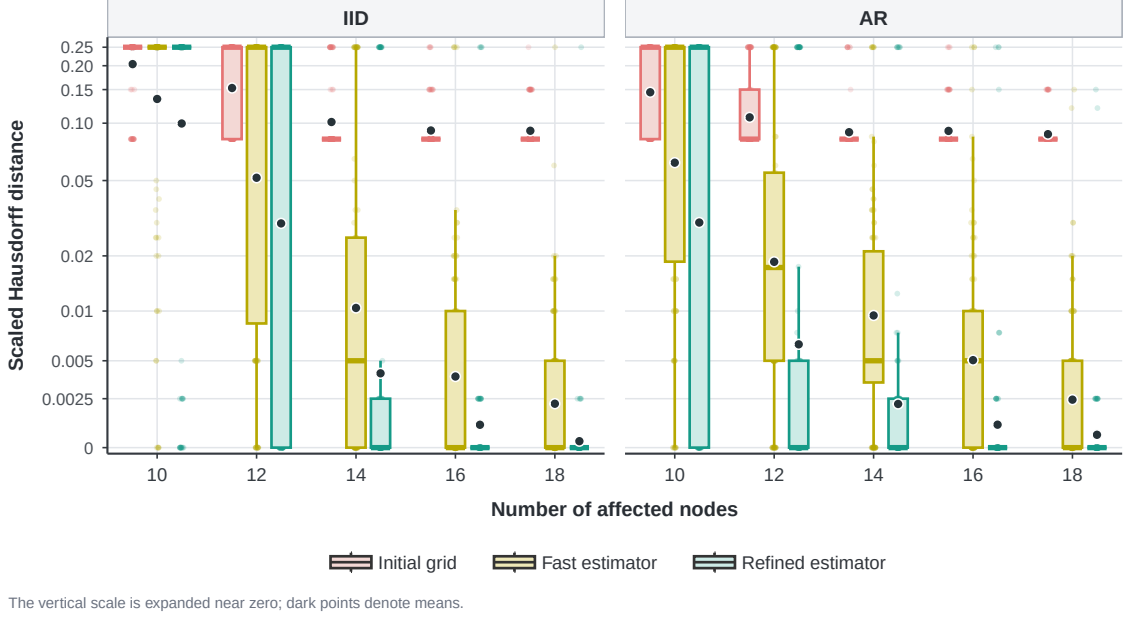


Figure 5: Scaled Hausdorff distances of the guessed, preliminary, and refined estimators across different signal strengths.

The gain from refinement becomes increasingly pronounced as the structural signal strengthens. When 18 nodes are affected, the refined estimator reduces the mean localization error by 97.2% and 96.6% relative to the guessed estimator, and by 55.1% and 50.0% relative to the preliminary estimator, under IID and AR observations, respectively. Under weaker signals, the additional gain from refinement is smaller because refinement adjusts the locations of detected change points but cannot recover a structural break omitted at the preliminary stage. These results illustrate the distinct roles of the two stages: the preliminary estimator recovers the break structure, while global refinement provides substantial additional localization accuracy once the complete change-point structure has been identified.

5 Macroeconomic Network Dynamics in FRED-MD

Data and empirical specification. We apply **GRIP** to the May 2026 vintage of the FRED-MD macroeconomic database, [available from the Federal Reserve Bank of St. Louis at the FRED-MD database website](#). The sample covers the period from January 1960 to April 2026, yielding $n = 796$ monthly observations. We retain $p = 50$ series spanning the major FRED-MD macroeconomic categories. Following the standard FRED-MD preprocessing scheme, each series is transformed according to its prescribed transformation code and standardized over the full sample period. Let $\mathbf{X}_t \in \mathbb{R}^{50}$ denote the resulting vector of macroeconomic indicators in month t .

We assume that the contemporaneous precision matrix of $\mathbf{X}_t$ is piecewise constant over time.

Accordingly, each estimated segment corresponds to a distinct regime characterized by a stable **contemporaneous precision-network structure**. The complete list of selected series, together with their category classifications, transformation codes, and additional implementation details, is provided in the Supplementary Material.

An important empirical feature of the FRED-MD data is the serial dependence that remains after the standard transformations. To illustrate this feature, Figure 6 reports the sample autocorrelations of the selected series after removing their estimated segment-specific means. Non-negligible autocorrelation persists at both short and moderate lags, providing descriptive evidence that temporal dependence is relevant for the empirical analysis and should be accommodated in both structural-break estimation and break-date inference.

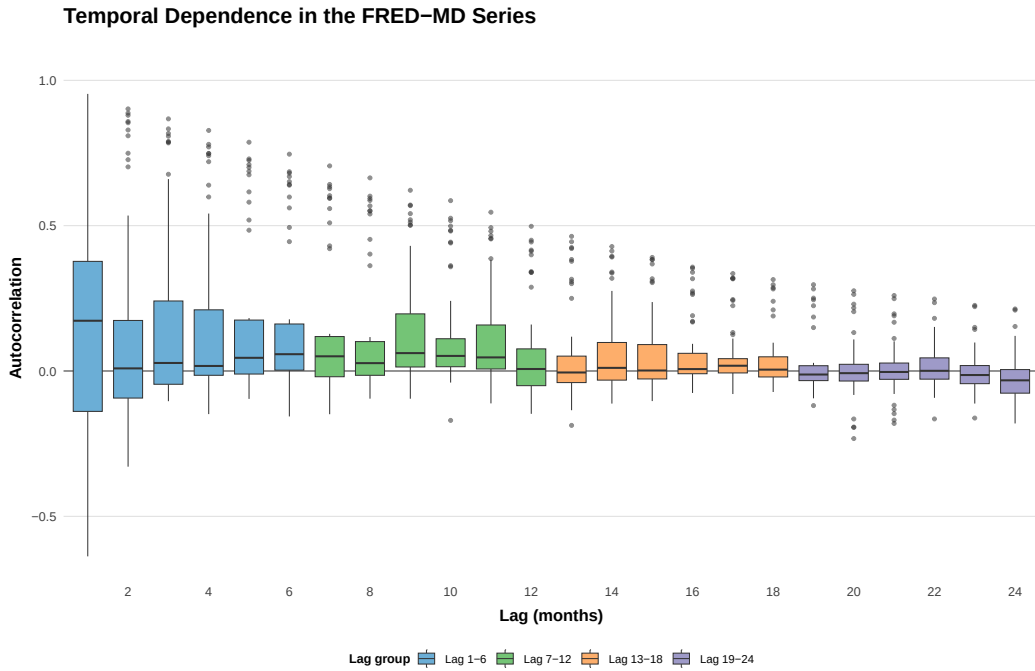


Figure 6: Sample autocorrelations of the transformed FRED-MD series after removing estimated segment-specific means.

Estimated structural breaks. Our proposed procedure identifies six changes in the **macroeconomic precision matrix based-network**, with refined estimates at June 1962, May 1975, March 2002, May 2008, October 2013, and April 2020. Figure 7 displays these estimates and their 95% confidence intervals. The reported break dates and confidence intervals are based on the globally refined estimates. **The first principal component is shown only to provide a low-dimensional visualization of the macroeconomic panel; the change points themselves are estimated from structural changes in the precision matrix.**

The confidence intervals reveal substantial heterogeneity in the temporal precision of the

estimated breaks. The changes in June 1962 and March 2002 are sharply localized, with intervals spanning only six months. By contrast, the intervals surrounding May 1975 and April 2020 span 21 and 18 months, respectively. This heterogeneity is important for economic interpretation: narrower intervals permit more precise temporal associations with historical events, whereas wider intervals call for greater caution in attributing a network reorganization to a particular episode.



Figure 7: Refined FRED-MD change-point estimates and 95% confidence intervals, with local views around the 1975, 2008, and 2020 changes.

Three of the six estimated breaks are temporally close to major episodes in U.S. macroeconomic history. The May 1975 point estimate occurs two months after the NBER-dated trough of the 1973–1975 recession, placing it near the transition from recession to recovery. This period followed the 1973–1974 oil-price shock and was characterized by elevated inflation and broad weakness in economic activity (??). Given the relatively wide confidence interval around the 1975 estimate, however, this correspondence should be interpreted as a broad temporal association rather than a precise dating of the economic transition. The May 2008 estimate falls in the early phase of the Great Recession, which began in December 2007, when declining house prices and mounting strains in mortgage and credit markets were increasingly affecting the broader economy (??). The April 2020 point estimate coincides with the NBER-dated trough of the COVID-19 recession and follows the World Health Organization’s characterization of COVID-19 as a pandemic in March 2020 (??). The wider confidence interval around the 2020 estimate similarly indicates that the precise timing of the associated network reorganization remains uncertain despite the close correspondence of the point estimate with the pandemic recession.

Network reorganization across regimes. Figure 8 compares the three most recent estimated regimes. The upper panels display the variable-level **precision matrix based-networks**, while the lower panels summarize connections at the macroeconomic-category level.

The macroeconomic network becomes progressively denser across the three regimes. The number of estimated variable-level edges increases from 37 in 2008–2013 to 42 in 2013–2020 and further to 55 after 2020. Cross-category connections also become more prevalent, indicating stronger interaction across different parts of the macroeconomy in the later regimes.

The variables that connect different macroeconomic blocks also change across regimes. In 2008–2013, the exchange-rate variable **EXCAUSx** is a prominent cross-category connector, linking real activity, financial-market, interest-rate, and price variables. During 2013–2020, **CLAIMSx**, a labor-market indicator, becomes an important bridge through its connections with monetary, credit, and financial variables. After 2020, **M1SL** assumes a more prominent cross-category role, connecting monetary conditions with real activity, labor-market, and price variables.

Overall, the estimated breaks are associated not only with increasing network density, but also with a reorganization of the variables that connect major macroeconomic sectors. This changing pattern of cross-category connectivity provides an economically interpretable view of the structural changes detected by GRIP.

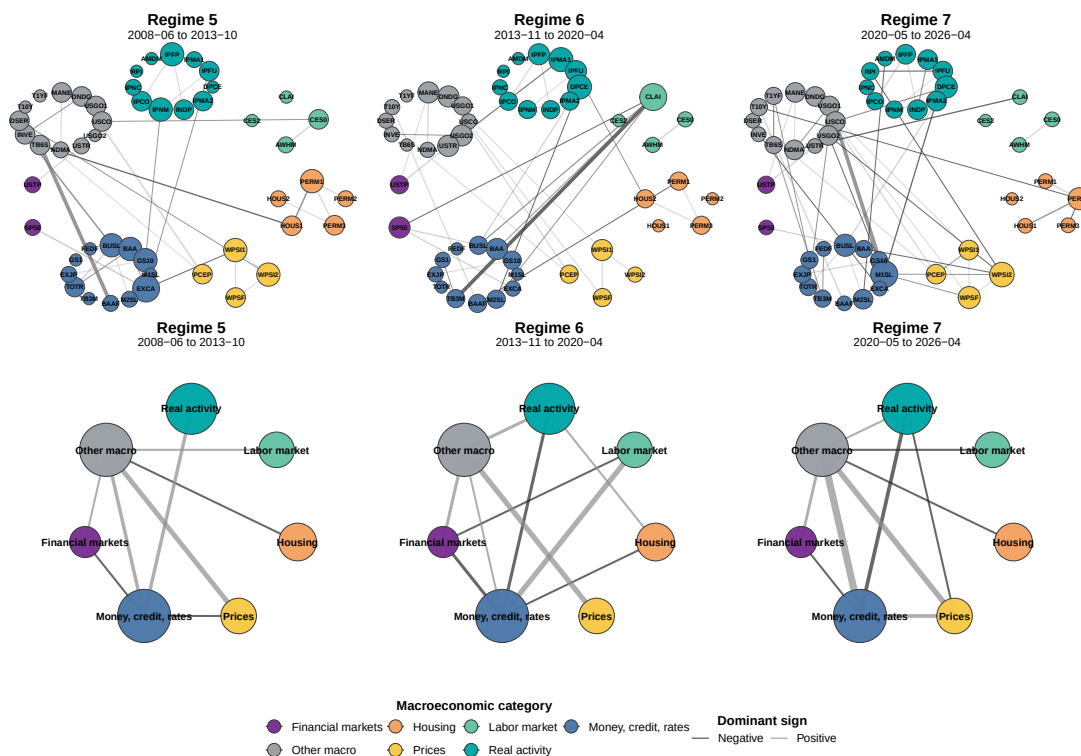


Figure 8: Variable-level and category-level FRED-MD conditional-dependence networks across the three most recent estimated regimes.

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A Unnecessary parts

The uniform eigenvalue bound in Assumption 2(a) also yields a fixed upper bound for the jump size. In particular, for every $j \in [k^*]$,

$$\kappa_j = \left\{ \frac{1}{p} \sum_{a=1}^p \|\boldsymbol{\theta}_{j+1}^{a*} - \boldsymbol{\theta}_j^{a*}\|_2^2 \right\}^{1/2} \leq \left\{ \frac{2}{p} \sum_{a=1}^p \left(\|\boldsymbol{\theta}_{j+1}^{a*}\|_2^2 + \|\boldsymbol{\theta}_j^{a*}\|_2^2 \right) \right\}^{1/2} \leq 2C_\rho. \quad (\text{A.1})$$

where the last inequality follows from the standard nodewise-regression coefficient bound implied by the uniform eigenvalue condition. The quantity $\sqrt{p} \kappa_j$ may be viewed as the Euclidean norm of the collection of nodewise jumps $\{\boldsymbol{\Psi}_j^{a*}\}_{a=1}^p$, with the corresponding group ℓ_1 norm given by $\sum_{a=1}^p \|\boldsymbol{\Psi}_j^{a*}\|_2$. The following elementary inequality will be used repeatedly in the proofs:

$$\sqrt{p} \kappa_j = \left\{ \sum_{a=1}^p \|\boldsymbol{\Psi}_j^{a*}\|_2^2 \right\}^{1/2} \leq \sum_{a=1}^p \|\boldsymbol{\Psi}_j^{a*}\|_2.$$

A.1 Intermediate results

We first recall the Bernstein inequality under temporal dependence and general tail behavior, which will be used repeatedly to control nodewise score fluctuations and segment-wise loss differences.

Lemma A.1 (Theorem 4 in Xu et al. (2024)). *Let $\{Z_i\}_{i \in \mathbb{Z}}$ be an $\mathbb{R}$ -valued, mean-zero, possibly nonstationary process with cumulative functional dependence measure $\Delta_{m,2}^Z$. Suppose there exist absolute constants $\zeta_1(Z), \zeta_2(Z), c_0, C_F > 0$ such that*

$$\sup_{m \geq 0} \exp\{c_0 m^{\zeta_1(Z)}\} \Delta_{m,2}^Z \leq C_F, \quad \sup_{i \in \mathbb{Z}} \mathbb{P}(|Z_i| > x) \leq \exp\{1 - x^{\zeta_2(Z)}\}, \quad x > 0.$$

Let $\zeta(Z) = \{1/\zeta_1(Z) + 1/\zeta_2(Z)\}^{-1} < 1$. Then there exist absolute constants $c_1, c_2 > 0$ such that, for any $x \geq 1$ and integer $n \geq 3$,

$$\mathbb{P} \left\{ n^{-1/2} \left| \sum_{i \in [n]} Z_i \right| \geq x \right\} \leq n \exp\{-c_1 x^{\zeta(Z)} n^{\zeta(Z)/2}\} + 2 \exp\{-c_2 x^2\}.$$

Lemma A.1 is repeatedly employed in proving deviation bounds for temporally dependent GGM processes. In the present paper, the combined parameter ζ in Assumption 1 plays the same role after applying the lemma to nodewise quadratic and cross-product sequences.

Lemma A.2 (Graphical parameter estimation). *Let $\hat{\boldsymbol{\Theta}}_j^g = ((\tilde{\boldsymbol{\theta}}_j^1)^\top, \dots, (\tilde{\boldsymbol{\theta}}_j^p)^\top)^\top$ and $\hat{\boldsymbol{\Theta}}_j^r = ((\hat{\boldsymbol{\theta}}_j^1)^\top, \dots, (\hat{\boldsymbol{\theta}}_j^p)^\top)^\top$ denote the guessed and refined nodewise Lasso estimators, computed with*

tuning parameters λ^g and λ^r , respectively. Suppose Assumptions 1, 2, and 4 hold, and let $\iota^* = \tilde{\iota} - (1 - 2\iota) > 0$. Under Assumption 3(a),

$$\max \left\{ \max_{j \in [k^*]} \max_{\tilde{j} \in (\mathfrak{m}(j-1), \mathfrak{m}(j)]} \frac{1}{p\kappa_j^2} \sum_{a=1}^p \|\tilde{\theta}_j^a - \theta_j^{a*}\|_2^2, \max_{j=0}^{k^*-1} \max_{\tilde{j} \in (\mathfrak{m}(j-1), \mathfrak{m}(j)]} \frac{1}{p\kappa_{j+1}^2} \sum_{a=1}^p \|\tilde{\theta}_j^a - \theta_j^{a*}\|_2^2 \right\} \leq C_2 \mathcal{I}_n^{-2\iota^*}$$

with probability tending to one. If Assumption 3 holds in full, then the refined estimators additionally satisfy

$$\max \left\{ \max_{j \in [k^*]} \frac{1}{p\kappa_j^2} \sum_{a=1}^p \|\hat{\theta}_j^a - \theta_j^{a*}\|_2^2, \max_{j=0}^{k^*-1} \frac{1}{p\kappa_{j+1}^2} \sum_{a=1}^p \|\hat{\theta}_j^a - \theta_j^{a*}\|_2^2 \right\} \leq \mathcal{I}_n^{-2}$$

with probability tending to one.

Lemma A.2 summarizes the two graphical estimation inputs used by the change-point analysis. The guessed estimators only need to be accurate enough for the preliminary reassignment, while the refined estimators are fitted on intervals localized by Lemma 3.2 and therefore converge at the stronger rate required for inference.